Society of Biomolecular Imaging & Informatics 2025

SBI2's 12th Annual Conference

Renaissance Boston Waterfront, Boston, MA, USA

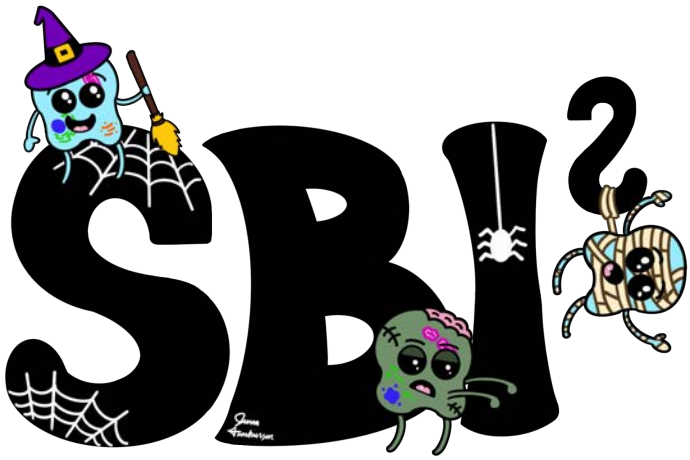
October 27 - October 29, 2025

Program & Events Guide



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THE 12TH ANNUAL SBI2 CONFERENCE

Sponsors

SBI2 recognizes the Sponsors of Biomolecular Imaging and Informatics 2025. The Society is grateful for their support.

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Araceli Biosciences

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Main Sessions & Keynote Spotlights

Lunch & Learn Sponsors

Sponsors Description

SBI2 recognizes the Sponsors of Biomolecular Imaging and Informatics 2025. The Society is grateful for their support.

YOKOGAWA	**Yokogawa Corporation** Yokogawa's High-Content Analysis systems offer advanced 3D live-cell imaging for drug discovery and biological research. Featuring confocal spinning disk technology and intuitive software like CellPathfinder, these customizable platforms enable high-throughput, automated analysis of spheroids, organoids, and live cells, delivering precise, rapid, and scalable phenotypic screening solutions.
MOLECULAR DEVICES	**Molecular Devices** We help scientists harness the potential of complex biology with next-generation technologies that accelerate discovery and support the development of new therapeutics. By combining advanced tools with scientific expertise, we empower researchers to generate deeper insights, solve pressing challenges, and drive innovations that improve human health worldwide.
revvity	**Revvity** Revvity provides health science solutions, technologies, expertise, and services that deliver complete workflows from discovery to development, and diagnosis to cure. Revvity is pushing the limits of what's possible in healthcare, with specialized focus areas in translational multi-omics technologies, biomarker identification, imaging, prediction, screening, detection and diagnosis, informatics, and more.
ARACELI BIOSCIENCES	**Araceli Biosciences** Araceli Biosciences is transforming therapeutic discovery with technologies built for speed, precision and impact. Our High Content Imaging platforms deliver real-time analysis and high-quality data that fuel AI-driven models. With instant insights today and predictive power tomorrow, we drive faster, more confident breakthroughs, helping patients access treatments sooner.
Cephla	**Cephla** Cephla offers scalable microscopy/imaging systems that give users the best of three worlds: access to latest technologies and application-specific optimizations from bespoke setups, flexibility of high-end research microscopes, and throughput and robustness of high content screening systems. Our open-source Python software gives users full control over integrations and workflow creation.
Glencoe SOFTWARE	**Glencoe Software** Glencoe Software builds and delivers innovative, scalable, easy-to-use scientific imaging solutions for its clients and partners. Our proven image database technology, OMERO Plus, makes the viewing, sharing, analysis and management of large sets of images and metadata easy and accessible to everyone in a group, team, project or organization.
SAGUARO Biosciences	**Saguaro** Saguaro offers data-rich, non-toxic fluorescent dyes for enhanced phenotypic profiling, cell viability, and cytotoxicity data. With its products and drug profiling service, Saguaro helps scientists around the globe with better, more reliable data. Saguaro non-toxic dyes include ChromaLIVE™ dye, allowing the richest image-based profiling, NucleoLIVE™ nuclear dye, MortaLIVE™ live-dead dye.
ramona	**Ramona Bio** Ramona's Vireo™ is the world's fastest and most cost-effective live-cell imaging system, marrying live-cell imaging with high-throughput screening. It is a compact, benchtop 24-plex multi-camera array microscope (MCAM®) for rapid cellular assessment (brightfield and four-

	channel fluorescent imaging in 2 minutes per plate) and novel video assay functionalities in multiwell plates.
Aurora Microplates	**Aurora** Aurora Microplates, a leader in microplate innovation, manufactures and supplies Cyclo-Olefin Polymer (COP) Micro-Titre plates to the Drug Discovery Sciences. With complete control over our supply chain from raw material to finished products, our microplates offer unparalleled clarity, flatness, and minimal optical noise, ensuring Optimal SNRs in all experimental conditions.
ZEISS inSphero	**Zeiss & InSphero** InSphero specializes in 3D cell culture and organ-on-a-chip technologies, advancing drug discovery, toxicology testing, and disease modeling for more accurate, predictive, and ethical research outcomes. Carl Zeiss Microscopy delivers innovative solutions for pharma R&D, featuring smart microscopy and scalable computing, enabling seamless transition from sample to insights for scientific discovery.
BROAD INSTITUTE	**Broad Institute** The Broad Institute Center for Discovery of Therapeutics (CDoT) and Imaging Platform have decades of experience pioneering cutting-edge imaging and image analysis technologies, and collaborating with academic and industry partners to run them at scale. Visit Booth 12 to learn about our new Optical Pooled Screening capabilities!
Genedata	**Genedata** Genedata transforms data into intelligence with innovative software solutions incorporating extensive domain knowledge. Leading biopharmaceutical organizations rely on Genedata to digitalize and automate R&D processes. From discovery to clinic, Genedata solutions help maximize the ROI in R&D. Founded in 1997, Genedata is headquartered in Switzerland with offices around the world.
Ardigen	**Ardigen** Ardigen is a AI Contract Research Organisation (CRO) dedicated to increasing the success rate of drug discovery. Recognized among the top 5% of AI in drug discovery companies worldwide, Ardigen integrates expertise in biology, bioinformatics and software engineering to deliver advanced computational solutions tailored for the biotechnology and pharmaceutical sectors.
Lumencor	**Lumencor** Lumencor advances research and development with illumination for clinical, life, and materials sciences. Light Engines, Imagers, and Scanners are tailored for application-specificity and manufactured for brightness, color selectivity, speed, stability, and precision. End-users, resellers, and OEMs benefit from lighting and photometric hardware supporting best quality data in clean, mercury-free laboratories.
curia	**Curia** Curia is a Contract Development and Manufacturing Organization. Our small molecule offering spans discovery to commercial manufacturing and fill-finish services, integrating scientific, process, regulatory and analytical capabilities for API life cycle. Our state-of-the-art facilities, scalable technologies, and commitment to growth advances your program through drug substance and drug product manufacturing.
Element Biosciences	**Element Biosciences** Element Biosciences is advancing drug discovery with AVITI24™, a next-generation multiomics platform. By integrating transcriptomic, proteomic, and morphologic data at single-cell resolution, AVITI24 reveals cellular function and disease mechanisms with unmatched clarity. Powered by Avidite™ chemistry, it delivers superior data quality, scalability, and cost efficiency to accelerate therapeutic development.
Agilent	**Agilent Technologies** Agilent, a global leader in analytical and clinical laboratory technologies, provides trusted answers that help scientists and clinicians solve scientific challenges, so they can bring great

	science to life.
CytoTronics	**CytoTronics**
etaluma microscopy simplified	**Etaluma** Etaluma's compact inverted microscopes are used inside incubators, hypoxia chambers, TC Hoods, and environmental control workstations with remote control and time-lapse image monitoring. Perform autofocus, tiling, stitching, z-stacks, and analysis for cell counting, confluence, and higher magnification custom assays in any microplate footprint labware. Python based software gives end-user flexibility
Mitsui Chemicals	**Mitsui Chemicals** Mitsui Chemicals was founded in 1912, with a philosophy of societal contribution through providing high-quality products and services with innovative material creation, harmonious with global environments. Our InnoCell cultureware is highly O2/CO2-permeable to improve cell function with low drug adsorption and high imaging quality, including for high-content imaging and organoids.



President's Welcome

Dear Colleagues and Friends,

Welcome to the 12th Annual Conference of the Society of Biomolecular Imaging and Informatics in Boston, MA. On behalf of the SBI2 Board members, councilors, and volunteers, I am delighted to welcome you to a gathering that celebrates discovery, collaboration, and the many connections that make our community strong.

This year's program reflects both the depth and diversity of our field. We begin with hands-on workshops, including a practical session on applying deep learning tools for segmentation and a career panel supporting the next generation of imaging scientists. The educational courses will guide us from the foundation principles of high content imaging and screening (HCI/HCS) through best practices in assay development, and real-world applications in drug discovery. Advanced tracks will highlight complex cellular models, 3D organoids, near-patient cell systems, and cutting-edge informatics for high-dimensional data visualization and analysis.

Our scientific sessions will showcase progress in multimodal and multi-omics data integration, demonstrate how Cell Painting is accelerating discovery, and present innovations in screening technologies through case studies from across the community. We are especially honored to welcome our keynote speakers: Dr. Anne Carpenter, who will share how image-based profiling and Cell Painting assay are reshaping drug discovery, and Dr. Marinka Zitnik, who will explore the emergence of "AI scientists" that integrate data and reasoning across biological scales to drive new biomedical insights. We will also have time to connect during poster sessions, reception, and conversations at the exhibit hall; moments that often become the most memorable and meaningful part of a conference!

SBI2 was founded on volunteer passion, generosity, and a commitment to advancing science while keeping it accessible and inclusive; that spirit continues today! This conference is possible thanks to the efforts of many who gave their time and energy to create a program that brings us all together to learn, exchange ideas, and support each other. Whether you are joining us for the first time or returning as part of the SBI2 family, I hope you find inspiration, new collaborations, and lasting friendships.

Thank you for being here! I look forward to the conversations we will share in the days ahead.

Kind regards,
Nasim Jamali, PhD
SBI2 2025 President

LUNCH AND LEARN

Applications of High-Content Screening in Compound Profiling, Target Inhibition, and Neuronal Health

Claudia McCown, PhD

UF The Herbert Wertheim UF Scripps Institute for Biomedical Innovation & Technology



Tuesday, October 28
12:45 – 1:45 PM

Room A

Join us!

Element Biosciences

AVITI24™ Integrated Single Cell Multiomic and Sequencing Platform

A more complete view of biology for tomorrow's breakthrough discoveries—today

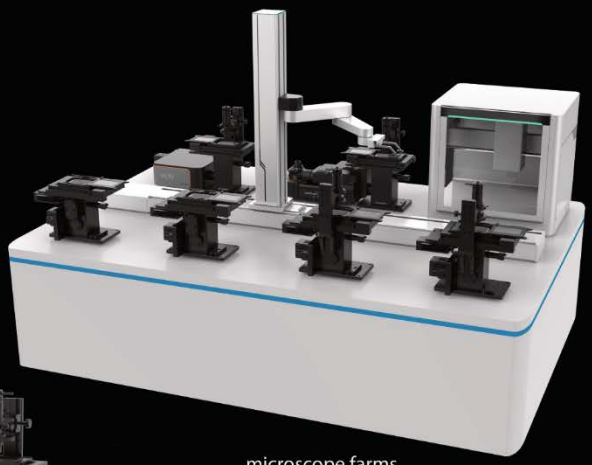
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for large scale data production



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About SBI2

The Society of Biomolecular Imaging and Informatics (SBI2) is an international community of leaders, scientists, and students promoting technological advancement, discovery, and education to quantitatively interrogate biological models to provide high-context information at the cellular level.

SBI2 Founders Message - a nod to history, State of the Society, and Welcome *Paul A. Johnston, Joe Trask, and Steve Haney*

Inaugural SBI2 High Content Cellular Imaging Symposium

North Carolina Biotech Center, RTP, NC, 11/10/2012



Left to right: Paul Johnston, Joe Trask, Mark Collins, Steve Haney

As founding members who participated in establishing SBI2 and leading its early development we've been asked to provide our perspective. SBI2 was boot-strapped by a grassroots movement of early adopters and experts in High Content Screening (HCS) technologies in the scientific community that routinely attended commercial scientific tradeshows or HCS specialized sessions at SLAS and ISAC (CYTO). Although these events helped highlight and bring the High Content community together it became apparent that the continued evolution of HCS would be better served by a professional society dedicated to highlighting the best and innovative science while providing educational resources to the community. In 2012 SBI2 was founded as a not-for-profit organization (501(c)3 status) with a culture grounded in the following core principles.

- **Volunteering & Affordability.** SBI2 was shaped by the passion of volunteer scientists and leaders who contributed their time and effort to emphasize high-quality cutting-edge science while maintaining affordability for the community of students, young investigators, and professionals.

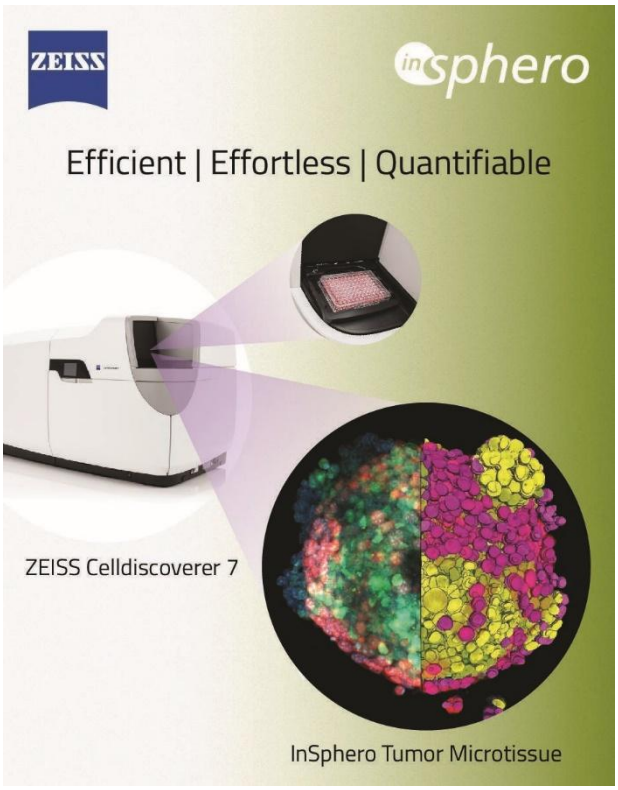
- **Rigor & scientific content**. In a rapidly evolving field encompassing multiple complex technologies (cellular technologies, microscopy principles, image analysis, assay validation, informatics, and data sciences), establishing and maintaining best practices was critical to advancing HCS science and its application to preclinical drug discovery. SBI2 addresses these needs through the educational programs and plenary sessions of its annual conferences, curated website content, and by peer-reviewed contributions to scientific literature.
- **Informatics.** When naming the society, initial emphasis was placed on biological aspects—hence terms like “Society of Biomolecular Imaging”. However, after some lively discussions (& a few beverages), we recognized the critical role of data analysis and added what has arguably become the most essential part of the SBI2 name: “informatics”. Since founding SBI2, informatics has evolved dramatically and now stands as a cornerstone of the entire HCS workflow.
- **Highlighting innovation**. SBI2 was founded with a commitment to drive innovation in the rapidly evolving fields of imaging and informatics. Our members actively champion the adoption and integration of emerging technologies, methodologies, and analytical approaches that push the boundaries of HCS. Through rigorous scientific exchange at annual conferences, publication of peer-reviewed research, and the development of accessible online resources, SBI2 serves as a catalyst for translating cutting-edge advancements into practical applications. This commitment ensures that SBI2 remains at the forefront of scientific discovery and technological progress.
- **Community & networking.** A founding strength of SBI2 is fostering collaborative networking alliances to advance science with global leaders of HCS technologies. These relationships have facilitated partnerships between academia, the pharmaceutical industry, and vendors, supporting career development for students and professionals in finding new positions or jobs and align with agencies to participate in “the standards” for the community.
- **Guidance & Educational Resources**. The SBI2 community offers guidance and educational materials to address many basic and fundamental questions in HCS. These include but are not limited to: how to design, develop, optimize and implement (automate) HCS assays; selection of appropriate HCS platforms and evaluation of what features or capabilities will be needed; considerations for using integrated image analysis software versus open-source image processing for data generation and analysis; best practices for applying appropriate statistical methods and quality control measures in HCS.
- **The Evolution of HCS.** The HCS field continues to evolve to more fully embrace and realize the “High Content” potential of these technologies. This progression includes the application of more complex cellular models, 3D biology, physiologically relevant primary cells, iPSCs, patient derived cells and organoids, spatial biology, digital histopathology, and the generation of big data from single cell analyses in HCS studies. Advanced microcopy techniques are being deployed to enable these analyses. Furthermore, image-based phenotypic and molecular profiling for drug discovery and toxicology are leveraging machine learning and AI-driven deep learning algorithms to analyze multiparametric data, address high dimensionality, and provide advanced visualization tools for interpretation.

A final message from the Founders. As founders, we take great pride in the remarkable growth and success of SBI2 and extend our deepest gratitude to the many individuals who have generously volunteered their time and expertise since its inception. Their dedication has been instrumental in shaping SBI2 while upholding its core mission and principles.

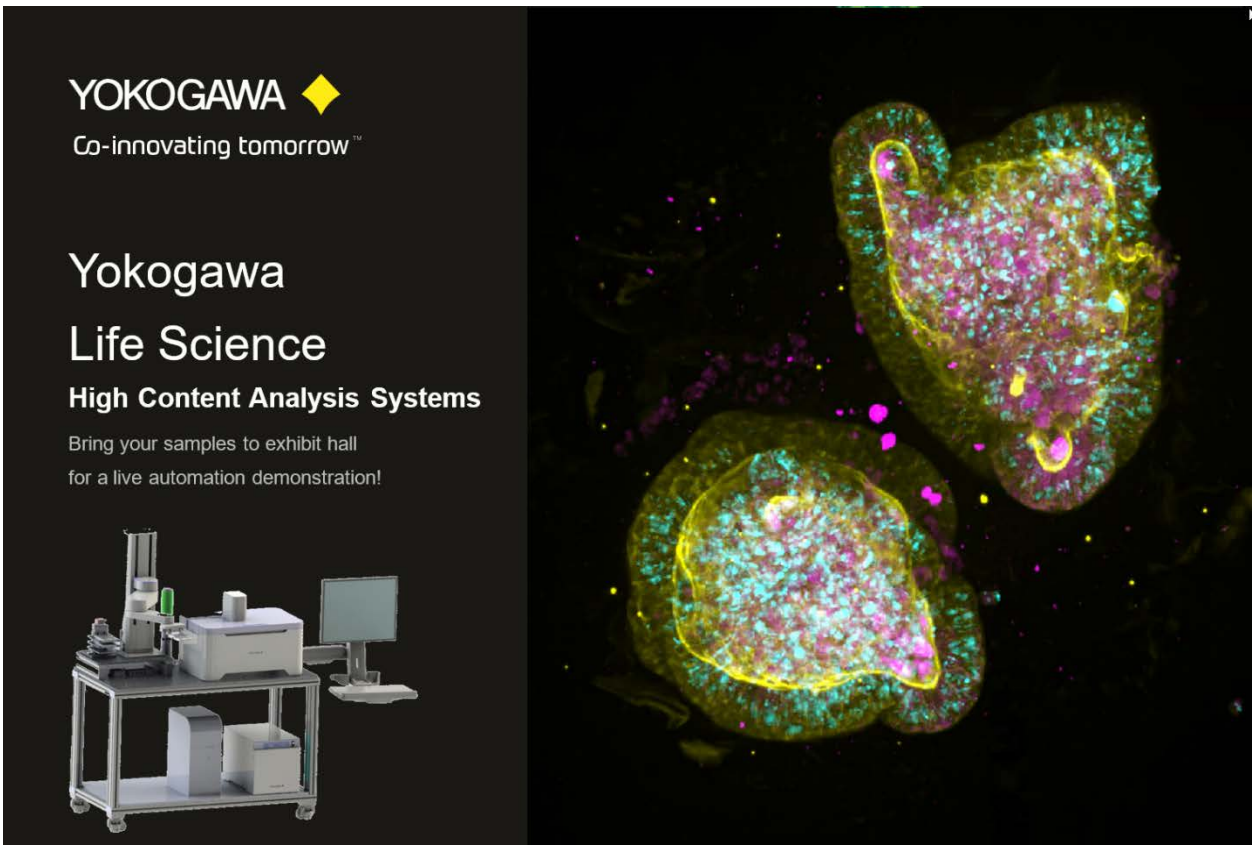
Looking ahead, we must remain steadfast in ensuring that SBI2 continues to deliver access to high-quality, innovative science at a reasonable cost, while embracing the diversity of disciplines and contributors that make our community so vibrant.

Based on the commitment of our volunteers serving on the SBI2 Board and committees, the thoughtful planning of the 2025 Annual Conference, and the exceptional quality of the scientific program and educational offerings, we are proud to affirm that “the state of the society is strong.”

We warmly invite you to join us and contribute to the continued growth and future success of the SBI2 family.


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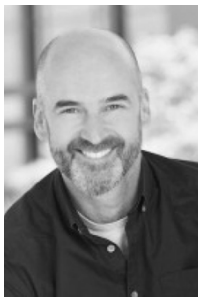
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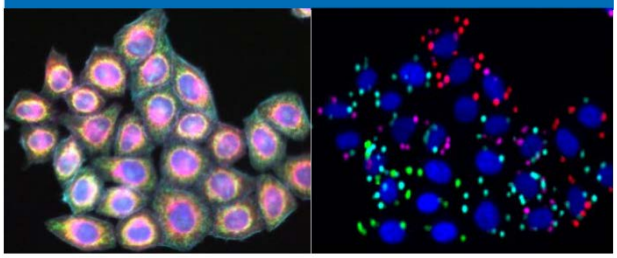
Joshua Bauer
Vanderbilt University

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Join our expert Clark Driscoll at Session 4: Screening Excellence to learn about the latest advancements in protein degraders research



Clark Driscoll
Senior Research Scientist; Cell Biology

Wednesday, October 29, 2:00-4:30pm
Renaissance Boston Waterfront Hotel
Booth #4

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Conference at-a-Glance

Monday, October 27th

7:30 AM	**Check-in opens**
9:00 AM – 12:00 PM	**Hands-on Workshops:**
Room: Salon G	● **9AM - 12PM Using Deep Learning tools for segmentation in CellProfiler** - Beth Cimini & Erin Weisbart (Broad Institute)
Room: Salon F	● **9 AM - 12PM Nextflow for Bioimaging** - Ken Brewer (Seqera)
10:30 AM - 11:45 AM	**Career Panel.** *Atlantic Ballroom*
12:00 PM – 1:00 PM	**Lunch**
1:00 PM – 4:30 PM	**Educational Courses**

- **Track 1: Introduction to HCI/HCS.** *Salon F*
  - **1-2 PM** Introduction & Overview of HCS/HCI/HCA
    - Steven Haney (SAH Consulting)
  - **2-3 PM** Best practices for HCS/HCI/HCA Assay Development & Optimization
    - O. J. Trask (Revvity)
  - **30 min Break (3-3:30 PM)**
  - **3:30-4:30 PM** An HCS Journey Along the Prostate Cancer Drug Discovery Path
    - P. A. Johnston (University of Pittsburgh)
- **Track 2: Advanced HCI/HCS Models.** *Salon G*
  - **1-2 PM** Complex Cellular Models
    - Jim Finley (Pfizer)
  - **2-3 PM** Imaging in 3D: Organoids & near-patient cell models
    - Brinton Seashore-Ludlow (Karolinska Institute)
  - **30 min Break (3-3:30 PM)**
  - **3:30-4:30 PM** Live Cell Bright Field Imaging & Analysis
    - Ilya Goldberg (ViQi Inc.)
- **Track 3: Advanced HCI/HCS Informatics.** *Salon H*
  - **1-2 PM** AI-based Image Analysis
    - Adriana Borowa (Ardigen)
  - **2-3 PM** Multiparametric data analysis, dimensionality & visualization methods
    - Jonny Sexton (University of Michigan) & Jenna Tomkinson (University of Colorado Anschutz)
  - **30 min Break (3-3:30 PM)**
  - **3:30-4:30 PM** Cytomining Ecosystem
    - Dave Bunten (University of Colorado Anschutz), Niranj Chandrasekaran (NYSCF), & Juan C. Caicedo (University of Wisconsin - Madison)

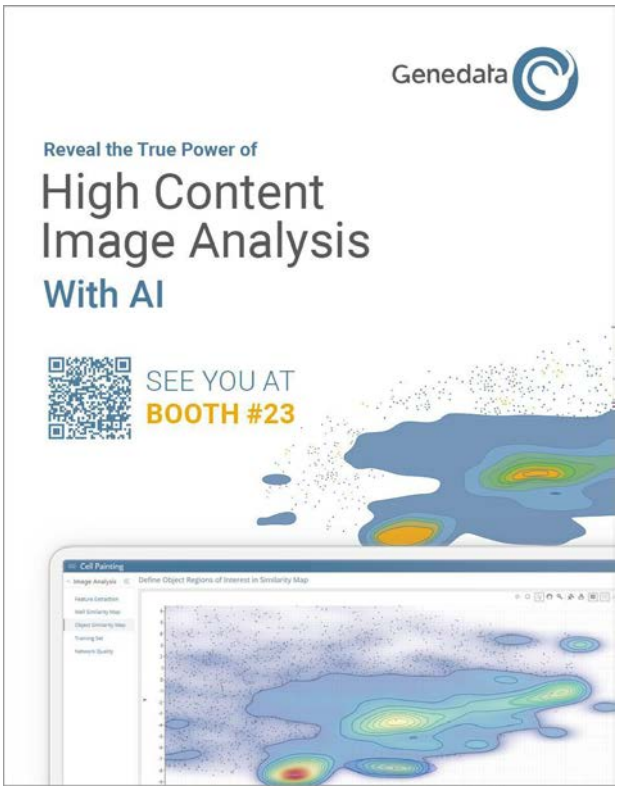
4:30 PM – 6:30 PM	**Post-Educational Courses Reception and Exhibit viewing**

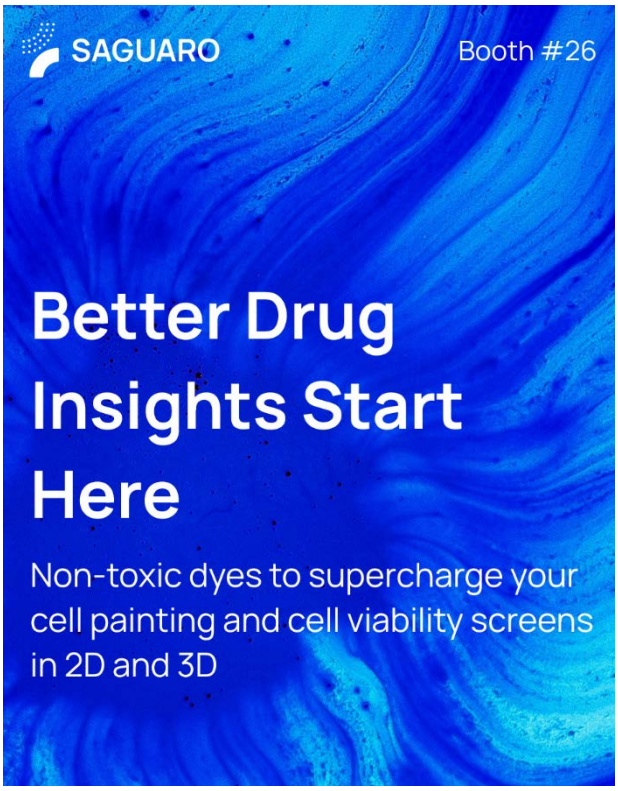
Tuesday, October 28th

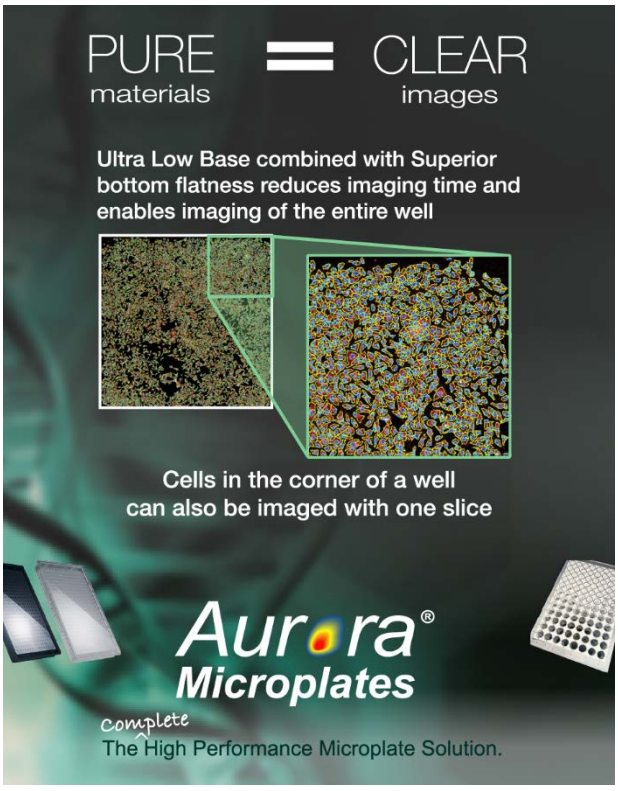
7:30 AM	**Check-in opens**
8:00 AM – 7:30 PM	**Exhibit Hall Opening**
	All keynotes and scientific sessions take place in the Atlantic Ballroom
8:50 AM – 9:00 AM	**Opening Remarks:** Nasim Jamali, SBI2 President.
9:00 AM – 10:00 AM	**Keynote Speaker:** Marinka Zitnik (Harvard).
	Empowering Biomedical Discovery with "AI Scientists"
10:00 AM – 10:15 AM	Exhibitor Spotlight - Molecular Devices
10:15 AM – 10:45 AM	**Coffee Break & Exhibit Viewing**
10:45 AM – 12:45 PM	**Session 1: Multi-omics and multimodal integration** ● Samouil Farhi (Broad Institute) ● Ralda Nehme (Stanley Institute) ● Stefan Semrau (NYSCF) ● Exhibitor Spotlight – Ardigen
12:45 PM – 1:45 PM	**Lunch & Learn Sponsored Sessions** ● Araceli Biosciences. *Salon G* ● Element Biosciences. *Salon F*
2:00 PM – 3:30 PM	**Session 2: Advancements in AI and Foundation Models (part 1)** ● Johnny Yu (Tahoe Therapeutics) ● Safiye Celik (Recursion) ● Chris Probert (Conflux)
3:30 PM – 4:00 PM	**Coffee Break & Exhibit Viewing**
4:00 PM – 4:45 PM	**Session 2: Advancements in AI and Foundation Models (part 2)** ● Lukas Fischer (Merck) ● Exhibitor Spotlight - Glencoe
4:45 PM – 7:30 PM	**Poster Session, Exhibit Viewing, & Reception**

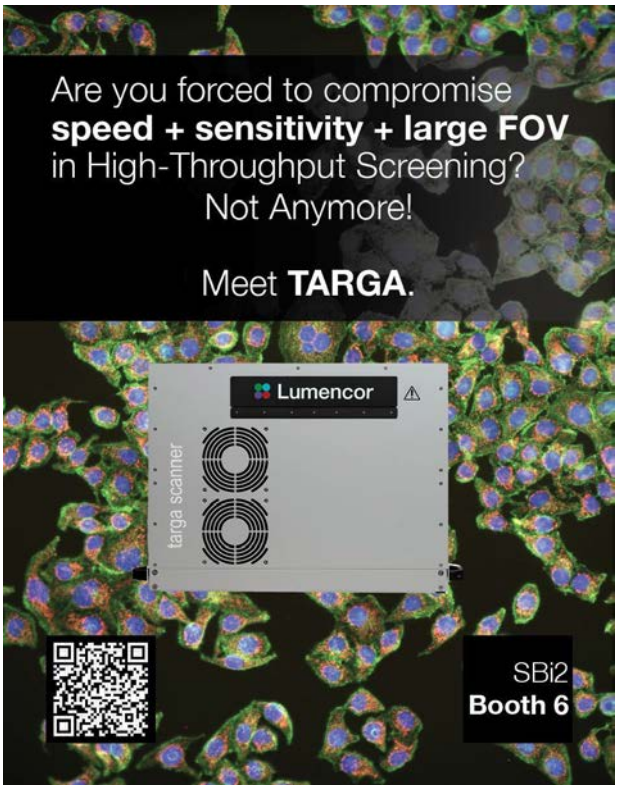
Wednesday, October 29th

7:30 AM	**Check-in opens**
8:00 AM – 5:00 PM	**Exhibit Hall Opening**
	All keynotes and scientific sessions take place in the Atlantic Ballroom
9:00 AM – 10:00 AM	**Keynote Speaker:** Anne Carpenter (Broad Institute)
	Cell Painting to accelerate drug discovery
10:00 AM – 10:15 PM	Exhibitor Highlight - Revvity
10:15 AM - 10:45 PM	**Coffee Break & Exhibit Viewing**
10:45 AM - 12:45 PM	**Session 3: Technological Innovations in Screening** ● Thomas Hofstetter (Insphero) ● Ophir Shalem (U Penn) ● Kaylene Simpson (PeterMac, Australia) ● Exhibitor Spotlight – Yokogawa
12:45 PM- 1:45 PM	**Lunch & Learn Sponsored Sessions** ● Zeiss and InSphero. *Salon F*
1:45 PM - 3:30 PM	**Session 4: Screening Excellence: Insights from Best Case Studies** ● Jaime Cheah (Broad Institute) ● Simon Mathien (U Montréal) ● Yue Qin (Broad Institute) ● Exhibitor Spotlight – Curia
3:30 PM – 4:00 PM	**Coffee Break & Exhibit Viewing**
4:00 PM – 5:15 PM	**Awards Ceremony and Selected Poster Presentations** ● President's Innovation Award ● Selected poster presentations ● Award Ceremony: Best Posters
5:15 PM - 6:15 PM	**SBI2 Annual General Meeting (AGM)** The AGM will cover the Society's current status, finances, and open Board positions. The public is encouraged to attend and join as members.
6:15 PM	**End of Conference**


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Program Agenda: Day 1

Hands-on Workshops & Education Sessions

Monday, October 27th	
TWO HANDS-ON WORKSHOPS **9:00 AM – 12:00 PM (noon)**	
Workshop I	
Room: Salon G 3rd Floor	*Using Deep Learning tools for segmentation in CellProfiler* **Led by Beth Cimini and Erin Weisbart**, Broad Institute

Abstract

Attendees will be introduced to CellProfiler Plugins and their use to integrate deep learning tools into CellProfiler pipelines. **Attendees should have some existing familiarity and/or experience with CellProfiler and will need to bring a laptop with CellProfiler and Docker Desktop installed**. Participants will learn to use CellProfiler Plugins to incorporate deep learning tools like Cellpose or StarDist into CellProfiler workflows to enhance image analysis capabilities.

Workshop II	
Room: Salon F 3rd Floor	*Nextflow for Bioimaging* **Led by Ken Brewer**, Seqera

Abstract

Nextflow is a powerful and flexible open-source workflow management system that simplifies the development, execution, and scalability of data-driven computational pipelines. It is widely used in omics data analysis including imaging informatics to automate complex analyses, making it easier to manage and reproduce large-scale data analysis workflows.

This training workshop is intended as a "getting started" course for students and researchers familiar with processing bioimaging data but completely new to Nextflow. It aims to equip participants with foundational knowledge and skills in three key areas: (1) How Nextflow simplifies complex data processing (2) Practical hands-on execution of an example bioimaging Nextflow pipeline using the command-line interface (CLI). (3) How to configure Nextflow to run pipelines in your HPC or Cloud environment.

By the end of the workshop, participants will be well-prepared for tackling the next steps in their journey to use Nextflow as a tool in processing imaging data. Additional study-at-home materials will be provided for them to continue learning skills such as developing Nextflow pipelines from scratch. **Attendees will need to bring a laptop and sign up for a GitHub account**.

10:30 AM - 11:45 AM	**Career Panel** Moderator: Kristin Halfpenny (Araceli)
Room: Atlantic Ballroom 3rd Floor	**Topic: Career transitions** **Panelists:**

	Steven Haney, David Bunten, Debra Nickischer, Judi Wardwell-Swanson, Ilya Goldberg, & Evgeny Shlevkov
12:00 PM – 1:00 PM	**Lunch**
EDUCATION SESSIONS **1:00 PM – 2:00 PM**	
Track I	**Introduction to HCI/HCS**
Room: Salon F 3rd floor	*Introduction & Overview of HCS/HCI/HCA* **Steve Haney,** SAH Consulting

Abstract

This discussion will present the landscape of topics one encounters as they get started in HCS and imaging assays. General topics that are relevant to biology and drug discovery, such as assay design, will be covered in passing to allow more time for topics unique to imaging studies. These include the types of imaging assays that can be considered, the mechanics of capturing images for quantification and the quantification process itself. Image generation topics will include introducing development of multi-channel fluorescent assays for cell signaling and health endpoints as well as general phenotypic assays. Specialized assays including FRET-based biosensors for unique cellular events and live cells assays for immune cell function will be discussed. Image analysis, taking a set of images and extracting data can occur on several levels. The most common method for data analysis is to leverage an image analysis application, either proprietary for the instrument used or open source. Feature selection and statistical analyses to generate results will be discussed. Complex analytical methods, including profiling of multiple features, deep learning quantification and emerging AI methods will be covered as well.

Track II	**Advanced HCI/HCS Models**
Room: Salon G 3rd floor	*Practical guide to 3D cell culture & Introduction to Complex In vitro Models & Microphysiological Systems* **Jim Finley** (Pfizer)

Abstract

Introduction to Complex In vitro Models (CIVM) and Microphysiological Systems (MPS). CIVMs including 3D tissues and MPSs, are becoming increasingly integrated into drug discovery processes for safety, efficacy and ADME evaluation. The recently established FDA roadmap for human-relevant in vitro technologies highlights the application of new approach methodologies (NAMs) to advance the predictivity of preclinical testing and positively influence the 3Rs efforts. This educational session will offer an overview of CIVM platforms, with an emphasis on their application in preclinical safety assessments. Additionally, the session will discuss the applications, advantages, and limitations of various in vitro NAMs, ranging from 2D monocultures and co-cultures to more complex 3D spheroids, organoids, and chip-based platforms. Selected case studies involving liver, gastrointestinal, pulmonary, and neural tissue models will be presented to illustrate practical applications.

Track III	**Advanced HCI/HCS Informatics**
Room: Salon H 3rd floor	*AI-based Image Analysis* **Adriana Borowa** (Ardigen)

Abstract

This session will cover image analysis using AI models: convolutional neural networks as well as transformer architectures. We will start with preparing the data, ensuring consistency, and understanding the key requirements that make an “AI-ready” dataset. We’ll cover essential image transformations:

normalization, scaling, alignment, etc. which enhance performance and assure result reproducibility. We'll perform feature generation using AI models available online and the feature interpretation as well as other down-stream applications. By the end of the session, attendees will be equipped with the practical knowledge and tools needed to apply AI-based analysis to their own high-content screening datasets. Additionally, after the session, participants will get access to an interactive notebook that walks through the entire pipeline, from data preprocessing to model inference and feature visualization, written in Python and compatible with PyTorch deep learning framework.

EDUCATION SESSIONS
2:15 PM – 3:15 PM

Track I	**Introduction to HCI/HCS**
Room: Salon F 3rd Floor	*Best practices for HCS/HCI/HCA Assay Development & Optimization* **O. J. Trask** (Revvity)

Abstract

This session is designed for HCS practitioners engaged in drug discovery, phenotypic screening, and toxicology. Participants will explore foundational principles of high-content screening (HCS) assay development, as well as advanced topics including 3D imaging, complex cell models, and phenotypic discovery strategies.We will examine both fixed and live-cell imaging approaches, highlighting their respective challenges and opportunities. The session will also cover the integration of artificial intelligence, machine learning, and deep learning (AI/ML/DL) in assay development and optimization. To support continued learning, we will reference the NIH/NCATS Assay Guidance Manual, a valuable resource for best practices in assay design and implementation.By the end of this session, participants will be able to: a) understand the principles and workflows in HCS, b) design robust, reproducible high-content assays, c) optimize imaging and analysis parameters for complex biological systems, d) identify and troubleshoot common challenges in assay development, and e) apply statistical and quality control metrics to ensure assay reliability.

Track II	**Advanced HCI/HCS Models**
Room: Salon G 3rd floor	*Imaging in 3D: Organoids & near-patient cell models* **Brinton Seashore-Ludlow** (Karolinska Institute)

Abstract

Advanced cell models, including organoids and patient-derived systems, are transforming drug discovery and translational research. These models often rely on 3D culture environments to preserve complex phenotypes and clinically relevant responses. In this instructional session, we will explore the design, implementation, and analysis of 3D high-content imaging (HCI) assays tailored for precision medicine and pharmaceutical discovery. We begin with practical guidance on assay development—covering culture formats, staining strategies, and miniaturization for screening. The session then moves into imaging techniques and computational approaches, highlighting state-of-the-art tools for image acquisition, segmentation, and quantification in complex 3D structures. Through real-world case studies, we will showcase workflows that combine biological insight with robust analytical pipelines. Special emphasis will be placed on overcoming challenges unique to 3D systems, including assay reproducibility, imaging depth limitations, and data complexity. We will also discuss emerging trends, such as machine learning for 3D image analysis, multiplexed readouts, and integration with patient-derived data for functional diagnostics. Whether you are new to 3D imaging or looking to advance your pipeline, this session will provide actionable insights and a future-focused perspective on high-content imaging in translational research

Track III	Advanced HCI/HCS Informatics
Room: Salon H 3rd floor	*Multiparametric data analysis, dimensionality & visualization methods* **Jonny Sexton** (University of Michigan) & **Jenna Tomkinson** (University of Colorado Anschutz)

Abstract

This session will demonstrate strategies and tools for high-content imaging data analysis following morphology profile extraction, with a focus on preprocessing, exploratory data analysis, and developing AI/ML-based scoring systems. It will cover open-source tools for analysis including preprocessing techniques like normalization, batch alignment/correction, correcting for plate positional effects, and single-cell quality control. We will then discuss dimensionality reduction and exploratory data analysis, such as unsupervised UMAP for visualizing cell clusters, PCA, Leiden Clustering, mutual information, and earth mover's distance. Semi-supervised methods like AutoML and XGBoost will be discussed for feature discovery via SHAP analysis. Guided examples will be provided for tasks including cellular classification and morphologic profiling.

EDUCATION SESSIONS
3:30 PM – 4:30 PM

Track I	Introduction to HCI/HCS
Room: Salon F 3rd Floor	*An HCS Journey Along the Prostate Cancer Drug Discovery Path* **P. A. Johnston** (University of Pittsburgh)

Abstract

The objective of this educational course is to highlight a case where high content screening (HCS) was the key component of the lead generation campaign. A brief introduction to prostate cancer will be provided, incidence, existing therapeutic options, and current unmet clinical need. The androgen receptor (AR) signaling pathway and its role in the development and progression of metastatic castration resistance prostate cancer (CRPC) will be briefly described together with evidence supporting the hypothesis that inhibitors or disruptors of AR protein-protein interactions (PPIs) with the TIF2 (SRC2) coactivator might lead to the development of effective CRPC therapeutics. The development, optimization and implementation of an AR-TIF2 PPI biosensor HCS assay and its implementation to screen 143,000 compounds in two formats will be described together with the confirmation and characterization of the HCS hits. The application of orthogonal assays, target engagement assays, molecular docking studies, and medicinal chemistry hit-to-lead optimization to uncover the allosteric modulator mechanism of action of three lead series will be described.

Track II	Advanced HCI/HCS Models
Room: Salon G 3rd floor	*Live Cell Bright Field Imaging & Analysis* **Ilya Goldberg** (ViQi Inc.)

Abstract

This minicourse will focus on techniques for processing brightfield images, and more generally, contrast techniques emphasizing cellular gross morphology as opposed to characteristics of specific molecular biomarkers. The course will cover some considerations in image acquisition, but is primarily targeted at analysis using AI and ML techniques to detect and quantify changes in morphology or general cellular phenotype to quantify target phenotypes, as well as compare relationships across diverse sets of induced phenotypes.

A great deal of the difficulty in analyzing these types of images arises from the lack of contrast emphasizing cell boundaries, which leads to difficulties in segmenting cells. We will discuss approaches to the segmentation problem and segmentation-free analysis techniques based on the premise that morphology and phenotype in cultured cells are characteristics not necessarily constrained to segmentable objects. The use of AI and ML techniques without assumptions about specific image content necessitates good experimental design to ensure that, for example, AIs are trained on appropriate controls.

The course will cover analysis from sampling sub-image regions (via segmentation or tiling), followed by either direct training of deep-learning models, or by featurization and use of feature-based classifiers. Semi-supervised and unsupervised techniques and their application will also be addressed. Finally, the course will address the interpretation of AI-generated results in an experimental biological context, with specific examples using quantitative end-point assays, timecourses, and phenotypic profiling. We will use specific biological examples of end-point assays and exploration of phenotypic distance techniques, as well as software examples including code and visualizations.

Track III	Advanced HCI/HCS Informatics
Room: Salon H 3rd floor	*Cytomining Ecosystem* **Dave Bunten** (University of Colorado Anschutz), Niranj Chandrasekaran (NYSCF), and Juan C. Caicedo (University of Wisconsin - Madison)

Abstract

In this session, you'll see how a modular ecosystem of open-source tools from the Cytomining organization such as DeepProfiler, Pycytominer, CytoTable, coSMicQC, and CytoDataFrame can be composed into reproducible, scalable workflows for image-based profiling. Together, these tools link every stage of the data lifecycle, from feature extraction and harmonization to quality control, visualization, and analysis.

Built on shared open standards and transparent design, this ecosystem exemplifies the practice of open-source science where software, data, and methods are openly developed, tested, and improved by the community. By promoting interoperability and reproducibility, these tools allow researchers to build upon one another's work, adapt workflows to new biological questions, and scale from small experiments to large imaging screens. The Cytomining organization tools demonstrate how collaborative, open infrastructure can accelerate discovery and make image-based profiling more accessible, transparent, and sustainable.

4:30 - 6:30	**Reception**

Program Agenda: Day 2

Main Sessions

Tuesday, October 28th

MAIN SESSION Day 1
Room: Atlantic Ballroom

KEYNOTE
9:00 - 10:00

Marinka Zitnik, Harvard
Title: *Empowering Biomedical Discovery with "AI Scientists"*

Abstract

We envision "AI scientists" as systems that reason skeptically, integrate multimodal data, and interface with experimental platforms to learn and innovate autonomously. In this talk, we describe AI models that operate at four biological scales and pave the way toward "AI scientists". At the molecular scale, ProCyon is a multimodal foundation model that predicts protein phenotypes from sequence and structure, executes zero-shot molecular function inference, and generates phenotypic descriptions for proteins in the dark proteome. At the cellular scale, PINNACLE fuses single-cell transcriptomic profiles with protein interaction networks to model drug effects and nominate therapeutic targets with cell type specificity. At the therapeutic scale, TxAgent performs real-time reasoning over trusted, verified resources and clinically validated insights spanning all drugs developed since 1939. At the patient scale, COMPASS predicts immunotherapy response across treatments and cancer types using concept bottlenecks grounded in mechanistic biology, informing early-stage trial design and patient selection. We validate model outputs with multi-institutional patient datasets, emulated clinical trials, and experimental studies in vitro and in vivo. As these systems gain capability and collaborate with researchers, they can generate testable hypotheses, design research plans, guide experimental evaluation, and open new paths for biomedical discovery.



Marinka Zitnik
Associate Professor at Harvard

Keynote Bio

Marinka Zitnik is an Associate Professor of Biomedical Informatics at Harvard Medical School, at Kempner Institute for the Study of Natural and Artificial Intelligence at Harvard University and at Broad Institute of MIT and Harvard. Zitnik investigates foundations of AI that contribute to the scientific understanding of medicine and therapeutic design, eventually enabling AI to learn and innovate on its own. Her research won best paper and research awards, including the Kavli Fellowship of the National Academy of Sciences, Kaneb Fellowship award at Harvard Medical School, NSF CAREER Award, awards from the International Society for Computational Biology, International Conference in Machine Learning, Bayer Early Excellence in Science, Amazon Faculty Research, Google Faculty Research, Roche Alliance with Distinguished Scientists, and Sanofi iDEA-iTECH Award. Zitnik founded Therapeutics Data Commons, a global open-science initiative to access and evaluate AI across stages of development and therapeutic modalities, and she served as the faculty lead of the AI4Science initiative.

10:00 - 10:15

MOLECULAR DEVICES

Exhibitor Spotlight – Molecular Devices
Angeline Lim

	Title: *Screen Smarter and faster with the new HCS.ai High-Content Screening System*

Abstract
High-content imaging is essential in drug discovery, but as assays grow more complex, researchers need faster, more powerful imaging and analysis tools to extract high-quality data efficiently. The ImageXpress® HCS.ai system is our latest high-content screening platform, designed for speed, flexibility, and ease of use. It delivers high-quality data across diverse imaging assays, featuring multiple confocal options, laser excitation, water immersion, and environmental control. The redesigned MetaXpress® Acquire software simplifies setup for new users while offering advanced customization for experts. Seamless integration with IN Carta® Image Analysis Software enables AI-powered insights, making HCS.ai a powerful tool for translational research and high-throughput screening.

10:15 - 10:45	**Coffee Break & Exhibit Viewing**
Session 1 10:45 - 12:45	**Multi-Omics & Multi-Modal Integration** **Chair: Evgeny Shlevkov (Johns Hopkins University)**
10:45 - 11:15	**Ralda Nehme,** The Broad Institute of MIT and Harvard **Title:** *Unraveling the complexity using genetically diverse human cellular models*

Abstract
Growing evidence suggests that human genetic variation associated with complex traits—such as psychiatric disorders— disrupts communication between specific brain cell types. Yet, the nature of these interactions, and whether they represent shared mechanisms across diverse risk alleles, remains poorly understood. To address this, we use human induced pluripotent stem cells (iPSCs) to model neurons and astrocytes derived from genetically diverse individuals, along with targeted CRISPR technologies to study risk variants in the context of the human genetic background. I will discuss newly developed resources and assays that we established to measure genetic influences on living human cells. Using a combination of multiplexed and pooled approaches (such as cell villages) along with arrayed, non-pooled measurements, we have identified transcriptional signatures and morphological fingerprints linked to genetic and pharmacological perturbations. We found that cell-type specific gene expression programs that govern specific morphological features underlie astrocyte-neuron communications and are altered in the context of psychiatric disorders. Together, these findings underscore the power of multi-donor cellular models to reveal genotype-dependent cellular responses and nominate new avenues for therapeutic discovery.

11:15 -11:45	**Sami Farhi,** The Broad Institute of MIT and Harvard **Title:** *Spatial transcriptomics for in situ CRISPR screening*

Abstract
Pooled optical screens have enabled the study of cellular interactions, morphology, or dynamics at massive scale, but have not yet leveraged the power of highly-plexed single-cell resolved transcriptomic readouts to inform molecular pathways. Here, we present Perturb-FISH, which bridges these approaches by combining imaging spatial transcriptomics with parallel optical detection of in situ amplified guide RNAs. We show that Perturb-FISH recovers intracellular effects that are consistent with Perturb-seq results in a screen of lipopolysaccharide response in cultured monocytes, and uncovers new intercellular and density-dependent regulation of the innate immune response. We further pair Perturb-FISH with a functional readout in a screen of autism spectrum disorder risk genes, showing common calcium activity phenotypes in induced pluripotent stem cell derived astrocytes and their associated genetic interactions and dysregulated molecular pathways. Finally, we show that Perturb-FISH can identify neighborhood dependent perturbation effects in a complex tissue by showing immune-tumor interactions in a xenograft model engrafted with human PBMCs. Perturb-FISH is thus a generally applicable method for studying the genetic and molecular associations of spatial and functional biology at single-cell resolution.

11:45 -12:15	**Stefan Semrau,** New York Stem Cell Foundation **Title:** *Disease Modeling at the Intersection of AI, Automation, and Stem Cells*

Abstract

Advancements in drug discovery for complex diseases necessitates that cellular models are not only scalable and reproducible but also pertinent to the specific disease in question. To fulfill these needs, we have established an integrated platform that merges the fields of stem cells, automation, and artificial intelligence (AI).

By utilizing the NYSCF Global Stem Cell Array®, we have automated many aspects of stem cell biology, from iPSC generation to their differentiation and phenotyping. This platform has been pivotal to the generation of several large-scale iPSC biobank resources, including Parkinson's, Alzheimer's Disease, AMD, and Autism, enabling researchers around the world access to ready-to-use iPSC samples. In addition to high-throughput reprogramming, additional technologies such as genome editing have been platformed, allowing gene-editing and -tagging to occur across multiple lines in parallel.

Beyond developing the foundational resources for high-throughput biology, we have automated workflows that facilitate modeling of complex disorders across genetically diverse patient cohorts, providing a robust platform for disease modeling and drug discovery through a range of data-generating assays, including sequencing, high-content imaging, and deep learning. Our phenotypic profiling pipeline has the capability to extract rich morphological embeddings and features from patient-derived cells. Subsequently, we can train machine learning models that are capable of detecting disease signatures, capturing individual-level variation, and maintaining robustness across large batches and layouts, ensuring the accuracy and reliability of our results.

12:15 - 12:30 Ardigen	**Exhibitor Spotlight – Ardigen** Nasim Jamali **Title:** *Beyond Images: Translating Morphological Profiles into Bioactivity Insights*

Abstract

Morphological profiling assays such as Cell Painting have emerged as powerful, target-agnostic tools for capturing rich cellular phenotypes across drug discovery programs. Yet, the challenge remains in translating these complex data into actionable biological insights and tractable novel starting points.

To address this, Ardigen developed phenAID (phenotyping Artificial Intelligence Driven), a platform for morphological profiling and compound prioritization. Two modules are highlighted here: bioactivity prediction and virtual screening. The bioactivity prediction module links phenotypic signatures to functional outcomes such as compound activity, mechanism of action, and early toxicity signals, enabling the prioritization of compounds prior to experimental validation. The virtual screening module extends these insights by identifying new chemical matter predicted to reproduce a desired phenotype, supporting the design of focused libraries and exploration of broader chemical space.

Case studies with pharmaceutical partners demonstrate that this approach supports compound triage in complex or low-signal assays and reduces the need for extensive functional follow-up. Moreover, image-based virtual screening was shown to deliver up to a four-fold increase in scaffold diversity compared with structure-based screening.

In summary, phenAID bridges the gap between cell-image phenotypes and small-molecule design by combining morphology-based bioactivity prediction with phenotype-guided virtual screening, increasing hit confirmation rates, chemical diversity, and speed to actionable leads.

12:45 - 1:45	**Lunch & Learn Sponsored Sessions**

Room: Salon F Element Biosciences	**Element Biosciences** Michael Previte **Title:** *A Complete View of the Cell: Unified Morphology, RNA, and Protein Profiling with AVITI24*

Abstract

High-content imaging and computational analysis have revolutionized the study of cellular responses to genetic and chemical perturbations. Approaches such as Cell Painting demonstrate how morphological profiling can capture rich phenotypic fingerprints, yet these insights are often disconnected from complementary genomic modalities. While adding RNA and protein expression data alongside morphology has been shown to improve predictions of mechanism of action (MOA) and mechanism of toxicity (MOT), current methods require separate workflows and instruments, resulting in increased complexity and a loss of paired statistical power across modalities.

In this seminar, we will introduce AVITI24™, the first integrated platform to unify phenotypic, transcriptomic, and proteomic analysis within a single workflow at single-cell resolution. By capturing morphology, RNA, and protein features simultaneously, AVITI24 provides a more comprehensive view of cellular state and delivers stronger predictive power for MOA and MOT studies. Through case studies, we will demonstrate how this unified approach refines biological interpretation, accelerates discovery, and reveals cellular responses that remain hidden when modalities are measured in isolation.

Room: Salon G ARACELI BIOSCIENCES	**Araceli Biosciences** Claudia McCown **Title:** *Applications of High-Content Screening in Compound Profiling, Target Inhibition, and Neuronal Health*

Abstract

High-content screening (HCS) has become an essential strategy in drug discovery, providing multiparametric insight into compound activity and enabling both phenotypic and target-based approaches. A critical challenge in HCS is achieving the resolution and throughput required to interrogate biologically complex systems while maintaining scalability across diverse assay formats. Here, we present three case studies that illustrate how high-content imaging can advance drug discovery across different biological contexts. In the first, we optimized the Cell Painting assay to achieve robust morphological profiling in both 384-well and 1536-well formats on the Endeavor® platform, enabling unbiased characterization of compound mechanisms of action and toxicity. In the second, we developed and validated a live-cell, fluorescence-based assay to monitor arginyltransferase 1 (ATE1) activity, establishing a scalable platform for the discovery of selective inhibitors targeting this enzyme. In the third, we applied high-content analysis to induced pluripotent stem cell-derived neurons, using neurite outgrowth as a measure of neuronal health and screening natural product extracts for neurite outgrowth modulators. Combining high-resolution imaging with AI-based analysis enabled the detection of subtle morphological differences and the clustering of compounds according to their phenotypic effects. Together, these examples demonstrate the breadth of high-content screening applications, spanning morphological profiling, target-based drug discovery, and complex neuronal models. Advances in imaging throughput and resolution support the use of scalable, information-rich assays that can accelerate the identification and characterization of novel therapeutic candidates.

Session 2 2:00 - 4:45	**Advancements in AI & Foundation Models** **Chair: Brian Feng (Calico)**
2:00 - 2:30	**Johnny Yu**, Tahoe Therapeutics **Title:** *Fueling the Virtual Cell: Data as Infrastructure for Next-Gen Biology*

Abstract
Next-generation predictive models of human biology - the "virtual cell" - require reframing of data as foundational infrastructure. We present Tahoe-100M, our giga-scale atlas of >100 million single-cell transcriptomes from 50 cancer cell lines exposed to ~1,100 pharmacological perturbations. As the largest drug-perturbed single-cell dataset to date, Tahoe-100M was built using pooled "cell village" designs to minimize batch effects while scaling across thousands of conditions.
Our results show that foundation models trained on Tahoe-100M generalize to unseen drugs and cell types, enabling zero-shot prediction of transcriptional response. Leveraging the dataset, we identified and experimentally validated novel immunomodulatory compounds that upregulate MHC-I expression. Furthermore, we show that Saquinavir, a generic HIV protease inhibitor reported to have off-target effects in the cardiovascular system has novel mechanisms of action and targets deconvolvable only through datasets like Tahoe100M. We also demonstrate that integration of genome-wide perturb/crop-seq datasets and single-cell drug perturbation in matched models create a nexus that allows for target identification and transcriptome-guided drug discovery.
Tahoe-100M marks a shift from bespoke single-cell experiments to scalable, reusable data platforms. By combining scale, structure, and biological diversity, we demonstrate that foundational single-cell data can fuel the development of predictive and composable virtual cell models that accelerate discovery across therapeutic and mechanistic domains.

2:30-3:00	**Safiye Celik,** Recursion **Title**: *RxRx3-core: Benchmarking Drug-Target Interactions in High-Content Microscopy*

Abstract
High-content microscopy enables systematic analysis of cellular responses to chemical perturbations, making it a powerful tool for drug discovery. Yet, the development and comparative evaluation of computational methods for drug-target interaction analysis has been limited by the lack of standardized, publicly available benchmark datasets. To address this gap, we introduce RxRx3-core, a new benchmark dataset designed to accelerate the development and validation of machine learning and quantitative analysis methods in cellular biology.
The RxRx3-core dataset, available at rxrx.ai/rxrx3-core, provides a large, diverse collection of high-content microscopy images capturing morphological profiles of cells treated with a broad range of chemical compounds. The dataset is accompanied by computational tools and resources for benchmarking, available at github.com/recursionpharma/EFAAR_benchmarking.
In this talk, we will describe the design and generation of RxRx3-core, the latest in Recursion's series of publicly available datasets. We will trace the evolution of our modeling approach from early weakly supervised methods to the latest Phenom models. We will present a comparative evaluation of these models on the new benchmark, including OpenPhenom, our latest publicly released model. Finally, we will discuss future directions, highlighting how the RxRx3-core dataset, together with supporting benchmarking tools, provides a platform for novel analysis methods in high-content microscopy, advancing our understanding of cellular biology and accelerating drug discovery.

3:00-3:30	**Chris Probert,** Conflux **Title:** *Efficient and privacy-preserving quality assessment of digital pathology images through active learning and model distillation*

Abstract
Digital pathology requires robust quality control, yet manual assessment is time-consuming and subjective. We present a quality management platform that scales across stain types and artifacts to catch errant slides quickly, providing insights that allow labs to improve their underlying processes and drive error rates down over time.
To support label-efficient training, we built a few-shot pipeline around three steps: prototype, train, and distill. Class prototypes are first generated in a SAM-assisted labeling app that uses embedding search to

rank similar example candidates. We then train embedding-based segmentation models, using active learning to identify misclassified for manual labeling and to accept well-predicted positives for autolabeling. Finally, we use a model distillation step to produce an efficient, deployment-ready model.
We prioritize computational accessibility and privacy by creating models that can run efficiently on conventional hardware using ONNX runtimes, and by writing our inference and deployment stack in Rust, which allows targeting both edge (browser-based and native) and cloud environments. By operating within institutional boundaries we are able to significantly reduce latency, typically catching errant slides within seconds of images being available, all without transmitting data to external cloud resources.
Our results show generalization to multiple stain types (H&E, IHC, IF), artifacts (e.g., tissue preservation, stain oversaturation, and scan-related issues), scanner vendors and institutions. We believe our approach demonstrates the potential of combining modern ML techniques with practical deployment considerations to deliver clinically valuable AI tools that enhance pathology workflows and increase data quality and readiness for the future of diagnostic medicine

3:30 - 4:00	**Coffee Break & Exhibit Viewing**
4:00 - 4:30	**Lukas Fisher,** Merck **Title:** *Development and Evaluation of a Cell Painting Assay for Predicting Mitochondrial Toxicity in Drug Safety Assessment*

Abstract
Capturing cellular responses to chemical perturbations using unbiased phenotypic profiling assays, such as Cell Painting, offered a powerful approach for early in vitro drug safety evaluation. We aimed to develop a robust assay for assessing mitochondrial toxicity using Cell Painting and evaluated its suitability to be an integral part of a mitochondrial toxicity assessment framework First, we determined that U2OS cells could be leveraged to identify phenotypic profiles with mitochondrial toxicants when cells were cultured to induce predominantly OXPHOS mediated energy metabolism. Glucose supplementation to enable glycolysis could rescue perturbation phenotype, revealing phenotypic effects were associated with mitochondrial toxicity Next, we evaluated the dataset scale needed to train classifiers required for reliable mitochondrial toxicity detection, considering both the number of compounds and replicates. Our datasets comprised a wide set (94 compounds, 2 replicates), a deep set (26 compounds, 7 replicates), and a combined set (118 compounds, 7 replicates). Included were literature reported mitochondrial toxins, compounds with no known association with mitochondrial toxicity and internal compounds with potential mitochondrial toxicity signals in other in vitro assays. Classification performance improved markedly with increased replicates, with the largest dataset yielding the strongest results (F1-score: 0.71). Notably, classification based solely on cell count performed at chance level. Finally, we identified differences in phenotypic profile clustering when comparing to Seahorse oxygen consumption data on perturbation of OXPHOS. Our study provides a comprehensive evaluation of Cell Painting for mitochondrial toxicity prediction, demonstrating consistent performance and underscoring its potential for integration into drug development screening.

4:30 - 4:45 Glencoe SOFTWARE	**Exhibitor Spotlight – Glencoe** Emil Rozbicki **Title:** *OMERO Plus Hosted powered by Portal: A hosted and managed platform for end-to-end imaging workflows across domains*

Abstract
Research organizations face growing challenges in managing, processing, and sharing large and complex imaging datasets. OMERO Plus Hosted is a fully managed service designed to simplify image data management for organizations of any size. Now powered by Portal, OMERO Plus Hosted provides a robust

orchestration platform for scalable, end-to-end image-based assays at some of the world's largest organizations.

Whether you're working with high-content screening data, digital pathology images, or large-scale microscopy datasets, Portal streamlines the process of importing, converting, analyzing and managing your data in the cloud. With modular integration for the most popular tools in bioimage data analysis, you decide how your images are processed. Tools for launching and reviewing analysis results are fully web-based, enabling seamless remote workflows.

The data lifecycle is inherently complex, and Portal not only manages the data at each step, but also captures the complete path the data has taken from initial registration to advanced analysis. With derived analysis results stored and organized in OMERO Plus, your workflows become fully traceable and reproducible.

Ultimately, Portal's automation reduces manual effort, minimizes errors, and accelerates your imaging workflows.

4:45 - 7:30	**Poster Sessions, Exhibit Viewing, & Reception**

Program Agenda: Day 3

Main Sessions

Wednesday, October 29th

MAIN SESSION Day 2
Room: Atlantic Ballroom

KEYNOTE
9:00 - 10:00

Anne Carpenter, The Broad Institute of MIT and Harvard
Title: *Cell Painting to accelerate drug discovery*

Abstract

Cell images contain a vast amount of quantifiable information about the status of the cell: for example, whether it is diseased, whether it is responding to a drug treatment, or whether a pathway has been disrupted by a genetic mutation. In image-based profiling, often using the Cell Painting assay, we extract hundreds of features of cells from microscopy images and measure the similarities and differences in the profiles to reveal connections among diseases, drugs, and genes. These strategies are now used throughout the pharma industry and underpin companies such as Recursion and SyzOnc. Because Cell Painting is inexpensive and high-throughput, and compatible with barcode-based optical screens, we can carry out experiments at very large scale, yielding single-cell profiles for hundreds of thousands of samples through public-private consortia (PERISCOPE/JUMP - genetic/chemical perturbations, OASIS - liver toxicity prediction, VISTA - rare disease drug repurposing, NIH IGVF - disease phenotype discovery). Our public Cell Painting Gallery contains over 700 TB of images and extracted profiles. These experiments are uncovering new functions of genes and new relationships among them. Cell morphology is therefore a powerful data source for systems biology alongside molecular omics.



Anne Carpenter,
Institute Scientist at
The Broad Institute of
Harvard and MIT

Keynote Bio

Dr. Anne Carpenter is an Institute Scientist at the Broad Institute of Harvard and MIT. Her research group, co-led by Dr. Shantanu Singh, develops algorithms and applications for large-scale experiments involving images. Her open-source CellProfiler software, now developed by Dr. Beth Cimini, is used by thousands of biologists worldwide and her Cell Painting assay has been adopted throughout the pharma industry to accelerate drug discovery - including in two startups based on her research: Recursion and SyzOnc. She leads industry-wide consortia such as JUMP-Cell Painting, OASIS, and VISTA. Her PhD is in cell biology from the University of Illinois, Urbana-Champaign and her postdoc in high-throughput image analysis was at the Whitehead Institute for Biomedical Research and MIT's CSAIL (Computer Sciences/Artificial Intelligence Laboratory). Carpenter has been named an NSF CAREER awardee, an NIH MIRA awardee, and is listed in Deep Knowledge Analytics' top-100 AI Leaders in Drug Discovery and Advanced Healthcare and STAT News' 2025 STATUS list 50 leaders in healthcare.

10:00 - 10:15

revvity

Exhibitor Spotlight – Revvity
Stephen Straub
Title: *Increasing the Information Content of High Content Assays with Deep Learning-Enabled Brightfield Segmentation and Novel Reagents*

Abstract

High content analysis has become indispensable for drug discovery and cell biology research. However, the information content of many assays is limited due to spectral overlap of traditional fluorophores and

inefficiencies in image and data analysis. To assist researchers in overcoming these limitations, Revvity has developed tools that touch every aspect of the imaging workflow, from microplates to cloud-based informatics solutions. Here, we detail a workflow for higher-order multiplexing utilizing novel fluorescent reagents and deep learning-enabled brightfield segmentation. We also discuss the new data management capabilities of Harmony 5.3 and show how these features simplify and expedite data relocation.

10:15 – 10:45	**Coffee Break & Exhibit Viewing**
Session 3 10:45 - 12:45	**Technological Innovations in Screening** **Chair: Kristin Halfpenny (Araceli)**
10:45 - 11:15	**Thomas Hofstetter,** InSphero AG **Title***: Machine learning-based, label-free detection of anti-fibrotic effects in a 3D retinal fibrosis model using bright-field imaging*

Abstract

Retinal fibrosis is a severe pathological process involved in several retinal diseases, including age-related macular degeneration (AMD). AMD is the leading cause of vision loss worldwide, projected to affect 288 million individuals by 2040. We have developed a 3D spheroid in vitro model based on primary human retinal pigment epithelial (hRPE) cells that recapitulate key features of retinal fibrosis, such as epithelial-to-mesenchymal transition (EMT) and aberrant extracellular matrix deposition. While these phenotypes are typically assessed using molecular or fluorescence-based assays, such approaches are labour-intensive, perturb the tissue, and limit scalability. In contrast, brightfield (BF) microscopy provides a rapid, inexpensive, and label-free readout that directly captures morphological alterations. We observed fibrosis-associated structural changes in spheroids that are readily detectable in low-resolution BF images. To exploit this, we implemented an automated image analysis pipeline combining segmentation, morphological and texture feature extraction, with a supervised machine learning model for fibrosis classification. The binary classifier achieved 91.6% accuracy at the single microtissue level, and 95.9% at the treatment condition level, distinguishing between fibrotic and anti-fibrotic responses.

Importantly, the method is compatible with spheroids cultured in 384-well plates, enabling high-throughput, non-invasive screening. Together, this work establishes BF-based machine learning as a scalable endpoint for phenotypic classification of fibrosis in a 3D retinal fibrosis model.

11:15 -11:45	**Ophir Shalem,** University of Pennsylvania **Title:** *Pooled gene tagging and in situ sequencing for deep profiling of subcellular proteome dynamics*

Abstract

System-level understanding of proteome organization and function requires methods for direct visualization and perturbation of proteins at scale across conditions and cell types. I will describe our labs efforts to develop scalable pooled non-disruptive protein tagging followed by in-situ barcode sequencing and image analysis and recent applications including small molecule mediated protein miss-folding, induction of ectopic protein protein interactions and imaging of sub cellular protein dynamics in response to stress.

11:45 - 12:15	**Kaylene Simpson,** Peter MacCallum Cancer Centre **Title:** *High-throughput patient-derived 3D tumor spheroid assay for CAR-T therapy pre-clinical evaluation*

Abstract

Conventional two-dimensional (2D) in vitro models fail to recapitulate the complex tumor microenvironment, limiting their utility in predicting CAR-T cell efficacy. There is a critical need for advanced preclinical models for solid tumors that accurately reflect CAR-T interactions.

We developed a novel in vitro assay in 384 well format, utilizing patient-derived three-dimensional (3D) colorectal adenocarcinoma spheroids to evaluate Lewis-Y-targeted CAR-T therapy. Utilizing high-throughput imaging, we quantified spheroid object-level CAR-T cytotoxic kinetics as well as endpoint killing dynamics that correlated with Lewis-Y- antigen expression. Imaging-based readouts demonstrated superior sensitivity compared to conventional bulk endpoint assays such as CellTiter-Glo. To establish mechanistic insights into CAR-T cell activation and distinguish it from untransduced T cells and healthy colorectal controls, we performed in situ, 384-well barcoded high throughput transcriptomics using Multiplexed Analysis of cells (MAC-seq) methodology and Flow cytometry to detect activation markers. Pathway enrichment analysis revealed distinct mechanisms of action for CAR-T cells.
Our 3D patient-derived tumor spheroid assay, integrated with imaging kinetics, MAC-seq transcriptomics, and cytometric profiling, offers a robust and physiologically relevant multi-OMICS platform for comprehensive CAR-T therapy evaluation. This multi-modal approach provides superior insights into CAR-T efficacy and mechanisms of action, making it a powerful tool to accelerate the discovery and development of next-generation CAR-T immunotherapies.

12:15 – 12:30	**Exhibitor Spotlight – Yokogawa**
YOKOGAWA	**Michael Lippincott**
	Title: *Live Cell Painting paired with time-lapse image-based profiling for predicting apoptosis*

Abstract

Cell death is a dynamic cell process that changes with time. The dynamics of single cells undergoing cell death cannot be understood from a snapshot in time. Instead, using live-cell time-lapse microscopy best allows the observation and quantification of how single cells' states change over time. Here we show the ability to quantify the onset and progression of one form of cell death, apoptosis, through time using live Cell Painting and time-lapse image-based profiling. We show that cells undergoing apoptosis can be discerned from cells not undergoing apoptosis over time. We also show that apoptosis can be predicted as early as 100 minutes after chemical induction. This technical advancement in time-lapse image-based profiling paves the way for future studies to better understand other cell death cell dynamics and how cells become poised for cell death. Understanding cell death dynamics is one piece to a larger puzzle, applicable to understanding therapeutic resistance, autoimmunity, cancer, and other diseases.

12:45 - 1:45	**Lunch & Learn Sponsored Sessions**

Room: Salon F

Zeiss & InSphero

Molly McQuilken & Özlem Yavaş Grining

Title: *From Microtissues to Insights: Enhancing Drug Discovery with InSphero and ZEISS*

Abstract

The FDA's plan to phase-out animal testing in preclinical research has led to a significant shift toward using complex in vitro models (CIVM) for drug testing and disease research. InSphero plays a pivotal role by providing microtissues that facilitate the adoption of 3D tissue cultures in biotech and pharmaceutical research, streamlining the integration of these advanced models into drug discovery R&D. Meanwhile, modern microscopes are capable of capturing detailed 3D and 4D images of such CIVM, offering unprecedented insights. However, classical 2D high-content analysis (HCA) approaches struggle with effectively analyzing these complex images. To address this challenge, ZEISS offers the arivis software platform, which leverages AI for advanced 3D/4D HCA. This platform, combined with ZEISS's imaging capabilities, automated workflows, and scalable computation, assists researchers in creating automated,

robust, and reproducible assays. Together, InSphero and ZEISS are accelerating the drug discovery process by enhancing the use of CIVM in research.

Session 4 1:45 - 3:30	**Screening Excellence: Insights from Best Case Studies** **Chair: Jenna Tomkinson (University of Colorado Anschutz)**
1:45 - 2:15	**Jaime H. Cheah,** The Broad Institute of MIT and Harvard **Title**: *Collaborative Approaches in Multi-layered High Throughput Screening Campaigns*

Abstract

The Center for the Development of Therapeutics at the Broad Institute works with collaborators to move their unique biological insights closer to becoming actual therapeutics. Embedded in this platform is the Collaborative Screening group, an integrated team of experts that deploy a suite of capabilities and a state-of-the-art automation facility to support our collaborators high throughput screens. These projects vary in scale, scope, assay readouts and complexity, but share similarities in their core needs and techniques for management. In this talk, I will be highlighting two of the projects and how the Collaborative Screening team has helped to navigate these multi-dimensional, multi-year projects:

1. PCCMA: working together with Dr. Gregory Way at University of Colorado to build the Pediatric Cancer Cell Morphology Atlas, providing a platform for phenotypic drug screening using Cell Painting.
2. OASIS: The 'Omics for Assessing Signatures for Integrated Safety Consortium is made up of academic, government, NGO, and biotech partners, whose goal is to assess the combination of Cell Painting, transcriptomics and proteomics in a variety of cell models for safety assessment, using hepatotoxicity as test case.

2:15 - 2:45	**Simon Mathien,** Université de Montréal **Title:** *Live-Cell Painting at Scale: Enabling High-Throughput Profiling in Live Cells*

Abstract

The High-Throughput Screening (HTS) core of the Institute for Research in Immunology and Cancer, Université de Montréal, supports collaborative drug discovery with academic and industry partners, with an established role in screening and a growing focus on high-content and morphological profiling. Here, we present the development and implementation of a scalable, automated live-cell painting workflow, enabling large-scale temporal phenotypic profiling in drug discovery applications. Key technical adaptations include a modular automation architecture optimized for timing-sensitive protocols, parallel assay handling, and maintain environmental stability required for live-cell workflows. We further optimized the workflow to support staggered compound addition, maintain consistent cell health across early and late timepoints, and enable time-resolved imaging with minimal phototoxicity. We will share performance benchmarks and validation data highlighting the robustness and reproducibility of the platform across multiple conditions. Two case studies illustrate the platform's utility: (1) a phenotypic screen in the rare disease ARSACS, where live-cell profiling enabled the identification of novel hits and mechanistic insights into compound series from prior campaigns; and (2) a collaborative screen with Saguaro Biosciences, in which 3,000 compounds were profiled in parallel using live-cell and fixed-cell painting, generating a comparative reference dataset for live-cell morphology under screening conditions. Together, these results illustrate how live-cell painting can be integrated into high-throughput screening workflows to enable large-scale morphological profiling and support mechanistic exploration of diverse compound libraries under live-cell conditions.

2:45 - 3:15	**Yue Qin,** The Broad Institute of MIT and Harvard **Title:** *From pixels to pathways: mapping biological programs from cellular images*

Abstract
The cell is a multi-scale structure with modular organization across at least four orders of magnitude. By integrating protein fluorescent imaging and protein biophysical associations, we have created hierarchical maps of human cell architecture. The map, known as the multi-scale integrated cell (MuSIC), identified numerous novel subcellular systems with unknown functions. Recent technology developments in high-content genetic screening approaches allowed us to systematically explore the biological functions of subcellular systems. By perturbing cells in a high-throughput manner and reading out cellular properties quantitatively as transcriptomics, proteomics, and so on, high-content genetic screening opens up the possibility to systematically chart functional organization of human cells. Recent work has established pooled microscopy-based phenotypic profiling (Optical Pooled Screening) as a cost-effective and robust approach to quantitatively monitor various cellular responses to genetic perturbations. We have implemented a series of experimental innovations to further empower the wide-spread usage of OPS in different biological contexts, and paved the way for integrating imaging with other multi-modal data to unlock a physical-functional cell map.

3:15 - 3:30 curia	**Exhibitor Spotlight – Curia** Clark Driscoll **Title:** *Case Study: Building a Scalable Immunofluorescence Workflow to Accelerate Protein Degrader Discovery*

Abstract
Immunofluorescence (IF) assays offer rich, high-content datasets but are seldom used as primary screening tools in drug discovery due to challenges in scalability. At Curia, we developed a flexible IF workflow that expands seamlessly from a 4-target, high-precision primary screen to a 10-target lead optimization assay. This approach, built through innovative assay design and close client collaboration, delivers run-to-run adaptability aligned with medicinal chemistry priorities, enabling faster, more informed decisions in protein degrader discovery.

3:30 – 4:00	**Coffee Break & Exhibit Viewing**
4:00 – 5:15	**Awards Ceremony and Selected Poster Presentations**
4:00 – 4:30	**President's Innovation Award**
	Kelvin Groot, University of Zürich **Title**: *Highly multiplexed imaging uncovers combinatorial perturbation effects of small molecule treatments*

Abstract
Image-based drug profiling captures comprehensive biological effects of small molecule treatments for drug discovery applications such as small molecule target identification and mechanism of action (MOA) prediction. However, current imaging modalities for image-based drug profiling in single cells have limited information content for understanding molecular mechanisms of action. To address this, we conducted a 42-plex iterative indirect immunofluorescence imaging approach to profile 148 approved oncology drugs and 15 phenotypic reference compounds in human PC9 lung cancer cells. Imaged targets include readouts for cellular metabolism, DNA damage, innate immunity, signaling pathways, and (membraneless) organelles. The 42-plex panel outperformed a reduced 6-channel panel in weakly supervised single-cell drug classification (F1-scores: 0.96 vs 0.68 for single-cell image crops; 0.70 vs 0.20 for single-cell features). Feature importance analysis identified both known and novel phenotypic drug effects, confirming that the models capture biologically meaningful cellular responses. We additionally identified and genetically validated polypharmacological effects of the EGFR inhibitor osimertinib on the autophagy lysosomal

pathway. These results demonstrate that increased multiplexing generates more comprehensive and MOA-specific cellular phenotypes that meaningfully encode polypharmacological effects. When combined with genetic perturbations, this approach enables systematic identification and discrimination of on-target from polypharmacological effects, providing a powerful platform for small molecule target identification.

4:30 - 4:45	**Selected Poster Talk 1** Marcus Chin **Title:** *Harnessing high-content and super-resolution imaging to profile subcellular morphologies*

Abstract

Organelles undergo dynamic morphological changes depending on cellular cues, stress, genetic factors, or disease. For mitochondria, their structural complexity and disease-relevance have stimulated efforts to generate image analysis tools for describing mitochondrial morphology for therapeutic development. Using high-content analysis, we measured multiple morphological parameters and employed unbiased feature clustering to identify the most robust pair of texture metrics that described mitochondrial state.

Here, we introduce a novel image analysis pipeline to enable rapid and accurate profiling of mitochondrial morphology in various cell types and pharmacological perturbations. We applied a high-content adapted implementation of our tool, MitoProfilerHC, to quantify mitochondrial morphology changes in i) a mammalian cell dose response study and ii) compartment specific drug effects in primary neurons. Next, we expanded the usability of our pipeline by using napari, a Python-powered image analysis tool, to build an open-source version of MitoProfiler and validated its performance and applicability. In conclusion, we introduce MitoProfiler as both a high-content-based and an open-source method to accurately quantify mitochondrial morphology in cells. We anticipate that this tool, combined with higher-resolution imaging modalities, can be applied to other relevant organelles such as lysosomes to facilitate mechanistic discoveries in cell biology and disease.

4:45 - 5:00	**Selected Poster Talk 2** Jiří Reiniš **Title:** *High-throughput vpCells phenotyping of global and target-specific effects of 105,000 compounds*

Abstract

Morphological profiling has emerged as a powerful imaging-based approach to capture complex cellular responses to chemical and genetic perturbations. We have previously established vpCells, a high-throughput method for endogenous multicolor protein fluorescent tagging, and a clone collection of over 1,000 labeled proteins in double-tagged cell lines. In a proof-of-concept screen, we combined high-throughput microscopy, computer vision and machine learning to detect specific subcellular protein localization changes in a vpCells pool spanning 61 proteins against 1,000 compounds, successfully validating this approach for target-specific phenotyping.

Here, we present an expanded platform that applies pooled vpCells screening for phenotypic annotation of a large compound library. We built a pool of 7 clones containing 14 endogenously tagged proteins alongside 8 fluorophores labeling key cellular compartments, including membranes, nuclei, nucleoli, endoplasmic reticulum, and mitochondria. This integrated system was profiled against 105,000 compounds encompassing chemical diversity libraries and approved drugs.

Our analysis yielded three major findings: First, we detected 226 protein-specific compound hits, demonstrating the sensitivity of our approach for identifying target-specific effects. Second, we performed a compound-centric analysis, observing compounds with similar effects clustering together in high-dimensional phenotypic space, enabling mechanism-of-action prediction. Third, for approximately 100 compounds we observed a highly distinct vacuolization phenotype, which we characterized through detailed follow-up investigations.

Our platform represents a novel hybrid high-throughput phenotypic screening approach, combining target-specific protein localization analysis with comprehensive cellular phenotyping for drug discovery applications. We also provide a companion website enabling visualization and interactive exploration of our dataset.	
5:00 – 5:15	**Award Ceremony: Best Posters**
5:15 - 6:15	**SBI2 Annual General Meeting (AGM)** The AGM will cover the Society's current status, finances, and open Board positions. The public is encouraged to attend and join as members.
6:15	**End of Conference**

Speaker Bios

Hands-On Workshops & Education Sessions



Beth Cimini
Associate Director, Bioimage Analysis,
The Broad Institute of MIT and Harvard

Dr. Beth Cimini is the Associate Director for Bioimage Analysis at the Imaging Platform at the Broad Institute in Cambridge, MA. The Cimini lab (est 2021) focuses on bioimage analysis tool creation (Piximi, Bilayers) and maintenance (CellProfiler), as well as on applying open source tools to novel biological problems; they collaborate with 20-30 outside labs a year (ranging from small academic labs through large pharmaceutical companies). She co-created and directs the Platform's Postdoctoral Training Program in Bioimage Analysis, and also leads the Broad efforts on community engagement and technology development for the Center for Open Bioimage Analysis (COBA). She is a proud recipient of both the Association of Biomolecular Resource Facilities Alan Smith "Mentor of the Year" award as well as BioImaging North America's Michael W Davidson award for contributions in "Inclusion, Education, Mentorship, Standardization, and/or Advocacy".



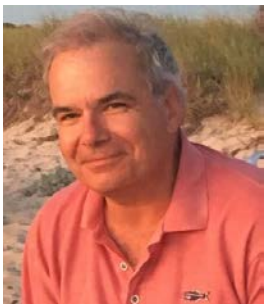
Erin Weisbart
Senior Computational Biologist,
The Broad Institute of MIT and Harvard

Erin Weisbart Ph.D. is a Senior Computational Biologist and the head of Image Analysis in the Cimini Lab, in the Imaging Platform at Broad Institute of MIT and Harvard. She is an expert in image analysis at scale, particularly the Cell Painting assay and pooled optical screening. She is lead curator of the Cell Painting Gallery, a public data repository, and contributes to CellProfiler software. Erin is particularly passionate about metadata and documentation.



Ken Brewer
Scientific Solutions Lead,
Seqera

Ken Brewer is a Scientific Solutions Lead at Seqera, where he supports the scientific community and customers in adopting modern software tooling and best practices. Prior to joining Seqera, he was one of the first computational biologists to join the early-stage startup ProFound Therapeutics, where he helped build out a computational platform for exploring the unannotated human proteome. He then joined GeneDx, where he provided technical leadership in building a bioinformatics platform designed to meet the data-intensive needs of the future of genetic medicine. Ken has a B.A. in Chemistry from Harvard College and a Ph.D. in Molecular Biophysics and Biochemistry from Yale University, where he developed computational methods for discovering structured noncoding RNA families in bacteria. He is passionate about open-source science and is a member of the cytomining and nf-core open-source projects, which are dedicated to providing high-quality software for analyzing microscopy images using machine learning and building bioinformatics pipelines, respectively.



Steven Haney,
Independent Consultant

Steven Haney has been involved broadly involved in drug discovery, from target identification to clinical development. Centrally focused on early drug discovery, Steven has led teams engaged in assay development, HTS and hit advancement, including extensive development of cellular assays for drug discovery and translational biology, working in HTS through Candidate Selection at Eli Lilly and translational biology in Oncology, Diabetes, Neurodegeneration and Cardiovascular Diseases at Eli Lilly, Pfizer and Valo Health.Steven has made several contributions to the Lilly/NIH Assay Guidance Manual on imaging assays, machine learning in morphological profiling and in fluorescent artifacts. He also held early roles at SBI2, including Secretary and President as well as Co-Organizer for the educational and plenary sessions for the first Annual Meetings of SBI2.



Paul A. Johnston. Ph.D.
Professor,
University of Pittsburgh

Dr Johnston has >35 years of drug discovery experience in the Pharmaceutical, Biotechnology, & academic sectors. An innovator of cell-based lead generation/optimization, he's a pioneer of the application of high content screening in drug discovery. In 2005, he designed/built the University of Pittsburgh Molecular Library Screening Center where he led 21 HTS campaigns & reconfigured the NCI 60 cell line assays for drug combination HTS. In 2011, he established independent chemical biology laboratories to apply novel drug discovery strategies to find new & effective drugs or combinations for prostate, melanoma, head & neck cancers, & hepatocellular carcinomas. He applies expertise in HTS/HCS assay development/implementation for drug discovery collaborations in other therapeutic areas. A member of SBS/SLAS since 1998, he is a founding member (2012) & president (2015) of the Society for Biomolecular Imaging & Informatics. Dr. Johnston teaches biochemistry & drug discovery in the Pharm.D. & Ph.D. programs.



Joe Trask
Life Sciences Segment Manager,
Revvity

Joe Trask is a member of Revvity's Life Sciences Segment, recognized for driving strategic customer engagement through teaching, training, and collaborations. With over 25 years of experience in drug discovery using cellular technologies - including high content screening, confocal microscopy, and flow cytometry - his work spans cancer, immunology, neurodegeneration, and toxicology. Joe is a published author, associate editor for the NIH/NCATS Assay Guidance Manual, and co-founder and inaugural President of the Society for Biomolecular Imaging and Informatics (SBI2).



Jim Finley
Principal Scientist,
Pfizer

Jim Finley is a Principal Scientist in the Drug Safety Investigative Toxicology group at Pfizer with 25 years' experience developing in vitro models supporting neuroscience drug discovery and preclinical toxicology. He specializes in development and implementation of complex in vitro models including co-cultures, Transwells, organoids, spheroids, organ-on-chips and iPSC's for safety and derisking endpoints in preclinical drug discovery.



Brinton Seashore-Ludlow
Senior Researcher,
Karolinska Institutet

Brinton Seashore-Ludlow holds a position Senior Researcher at Karolinska Institutet in Stockholm, Sweden. She is a team leader at the Department of Oncology-Pathology, where her research focuses on the application of functional precision medicine in ovarian cancer. In addition, she heads the assay development and screening team at the Chemical Biology Consortium Sweden.
Brinton received her PhD from KTH-Royal Institute of Technology in 2012. She then did a Postdoc in the lab of Stuart Schreiber at the Broad Institute of Harvard and MIT. Her work there focused on elucidating predictors of drug response in a large-scale cell line profiling dataset. Brinton then moved to the Chemical Biology Consortium Sweden at SciLifeLab. There she developed several high throughput adaptations of the cellular thermal shift assay (CETSA).



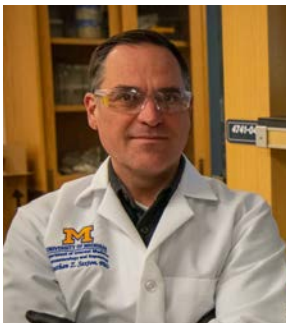
Ilya Goldberg
CSO,
ViQi Inc

Dr. Ilya Goldberg is the Chief Science Officer at ViQi where he leads the development of high-throughput imaging assays using AIs and machine learning technology. Prior to ViQi, Ilya co-founded a company that developed the first medical device to receive regulatory clearance for using an AI to help diagnose lung cancer in CT screening exams. Prior to this, Ilya led a research group at the NIH National Institute on Aging, where his group developed machine learning software for image processing in biology and medicine, and studied the molecular mechanisms of aging in humans and model organisms. Ilya earned his BS in Biochemistry from the University of Madison, his PhD in Cellular and Molecular Biology from The Johns Hopkins School of Medicine, and completed two postdocs in Crystallography at Harvard and in Image Informatics at MIT. While at MIT, Ilya co-founded the OME project which continues to be used for imaging infrastructure in large image repositories. He has over 60 peer-reviewed scientific articles in molecular and cell biology, pattern recognition, image informatics, and the basic biology of aging.



Adriana Borowa
Senior Data Scientist,
Ardigen

Adriana is a Senior Data Scientist at Ardigen responsible for the development of Ardigen phenAID platform that enables the identification of small molecule candidates. Her diverse experience includes work in digital pathology and neuron imaging with commitment to making meaningful contributions in various domains of life sciences. Adriana earned PhD in Technical Computer Science from Jagiellonian University and her research interests are focused on advancing the field of biomedical imaging by leveraging AI models. Her scientific work on cutting-edge deep learning algorithms aims to automate the analysis of High Content Screening images as well as microscopy images of bacteria.



Jonny Sexton
Associate Professor,
University of Michigan

Jonny is a chemical biologist at the University of Michigan who specializes in human organoid disease modeling, high-content screening, cell painting, and AI/ML and machine vision to enhance drug discovery. He is also the director of the U-M Center for Drug Repurposing, founded in 2019. He received his PhD in chemistry from the University of California, San Diego and moved to Stanford University as a postdoctoral fellow in Nobel Laureate Roger Kornberg's laboratory studying the structural biology of transcription complexes. He then moved to North Carolina and helped to start the BRITE institute and ultimately became the director of drug discovery. Jonny moved his laboratory to the University of Michigan in 2018 in the department of internal medicine, division of gastroenterology and hepatology and the department of medicinal chemistry. His research focuses on drug-induced liver injury and liver organoids as a platform for drug discovery.



Jenna M Tomkinson
Quantitative Cell Biologist,
University of Colorado
Anschutz Medical Campus

Jenna Tomkinson is a Quantitative Cell Biologist in the Gregory Way Lab at the University of Colorado Anschutz Medical Campus. She earned her Bachelor of Science in Biology with a minor in Behavioral Cognitive Neuroscience from the University of Colorado Denver in 2022. During her undergraduate studies, Jenna conducted research on the impact of temperature stress on gene expression across developmental stages in Drosophila melanogaster (common fruit fly). Jenna facilitates image analysis and image-based profiling for multiple projects in her current role, including studies on Neurofibromatosis Type 1 and pyroptotic cell death. She is also actively developing her skills in machine learning models to support various research initiatives at the lab, including studies on cardiac fibrosis and nuclear speckles. Jenna is committed to advancing the field of computational biology/bioinformatics and contributing to the development of innovative workflows and pipelines.



Dave Bunten, M.Ed.
Research Software Engineer
University of Colorado
Anschutz Medical Campus

Dave Bunten is a Research Software Engineer in the Department of Biomedical Informatics at the University of Colorado Anschutz Medical Campus, with over a decade of experience building sustainable software and including the Cytomining ecosystem of image-based profiling tools. He champions reproducible, community-driven bioimage informatics by creating accessible tutorials and resources that empower researchers to manage and analyze high-content imaging workflows. He holds an M.Ed. in Educational Technology and is passionate about mentoring and fostering lasting software stewardship.



Niranj Chandrasekaran
Data Engineer
New York Stem Cell Foundation

Niranj Chandrasekaran is a computational biologist who specializes in developing data pipelines and algorithms to solve complex biological problems. He is currently a Data Engineer at the New York Stem Cell Foundation, where he engineers reproducible, scalable processing pipelines for high-throughput cellular image analysis. He was, previously, a Postdoctoral Associate at the Broad Institute where he led a major consortium of pharmaceutical and non-profit partners to generate the world's largest publicly available Cell Painting dataset, a critical resource for drug discovery and building foundational models. His prior work in genomics and biophysics includes developing novel algorithms for identifying complex DNA structural variations and methods for predicting the conformational dynamics of proteins.



Juan C. Caicedo
Assistant Professor
University of Wisconsin–Madison

Juan Caicedo is a Morgridge Investigator and Assistant Professor at the University of Wisconsin–Madison. His research work is focused on the use of deep learning and machine learning to advance the understanding of cellular phenotypes and tissue organization in images. He started his independent research group as a Schmidt Fellow at the Broad Institute. Prior to this, we held postdoctoral positions in the Imaging Platform at the Broad, and in the Computer Vision group at the University of Illinois in Urbana-Champaign. He obtained his PhD from the National University of Colombia, and as a grad student he was a research intern at Google and Microsoft Research.



Debby Nickischer
Sr. IT Capabilities Manager, Biology
Arvinas

Debby is a seasoned Research Scientist with over 24 years of experience in high content imaging and screening. Currently, as Sr. Manager of IT Capabilities in Biology at Arvinas, Inc., she leads the integration and implementation of innovative IT solutions to streamline R&D operations. Her first 10 professional years were in clinical laboratory science focusing on Immunology and Virology. She then moved to Pharma where she built her expertise in high content screening, image analysis, flow cytometry, drug discovery assay development, HTS and automation, with a focus on advancing imaging platforms for drug discovery. She has also worked in industry as a FAS and contributed to developing new products. She has lead teams in clinical labs and research labs. Her leadership in these areas has developed her role as a key mentor.



Evgeny Shlevkov
Assistant Professor
Johns Hopkins University
School of Medicine

I am an Assistant Professor in the Department of Physiology, Pharmacology, and Therapeutics at Johns Hopkins and an Assistant Director of Drug Discovery at the Lieber Institute. My laboratory uses cellular phenomics to investigate the cellular and molecular mechanisms underlying schizophrenia. Before joining Hopkins, I designed and implemented the in vitro pharmacology strategy at Myrobalan Therapeutics, a CNS-focused small-molecule start-up. In that role, I oversaw a portfolio of candidates, from lead identification (Lead ID) to preclinical candidate (PCC), including molecules belonging to the kinase inhibitor, GPCR inhibitor, and Ion Channel Positive Allosteric Modulator (PAM) classes for schizophrenia and other indications. Prior to Myrobalan, I led pipeline and exploratory small-molecule programs at Biogen. My team was the first to discover a small-molecule activator of the Parkin E3 ligase, a breakthrough that opened the door to a first-in-class therapeutic for Parkinson's Disease.



Judi Wardwell-Swanson
Senior Scientist
InSphero

Judi Wardwell-Swanson is a Science and Technology consultant at InSphero and past-president of the Society of Biomolecular Imaging and Informatics (SBI2) board of directors. At InSphero her work is focused on the development of high content imaging applications utilizing 3D human microtissue models. Judi also has accumulated more than 25 years of experience in the pharma industry and has a long-standing interest in the utilization of genomics and imaging technologies to help drive drug discovery. During her tenure at Bristol-Myers Squibb, her team conducted phenotypic genetic and chemigenomic screens in physiologically-relevant cell models with an emphasis on the identification and validation of new drug targets. Judi is also the author of several journal articles and a book chapter on high content imaging-related topics including HCS data management, analysis of multidimensional imaging data, and 3D imaging and analysis of multicellular spheroids and micro-tissue models.

Main Session Day 1



Ralda Nehme
Director,
The Broad Institute of MIT and Harvard

Dr. Ralda Nehme is a Principal Investigator at the Stanley Center for Psychiatric Research at the Broad Institute of MIT and Harvard, where she also directs the stem cell program. Her lab investigates the genetic, cellular, and molecular mechanisms driving neurodegenerative, neurodevelopmental and psychiatric disorders, with a focus on how human genetic variation shapes cellular phenotypes. Combining human pluripotent stem cell models, genome editing technologies, high-content imaging, electrophysiology, and multiomic approaches, the Nehme lab aspires to build scalable and quantitative experimental approaches for dissecting disease-relevant biology.

Dr. Nehme earned her B.S. from the American University of Beirut and completed her Ph.D. at Dartmouth College, studying neuronal development in C. elegans. She conducted postdoctoral research at Harvard University, where she developed robust methods for generating human stem cell–derived neural cells and became engaged in modeling psychiatric disease.



Sami Farhi
Director, Spatial Technology Platform,
The Broad Institute of MIT and Harvard

Sami Farhi leads the Spatial Technology Platform at the Broad Institute of MIT and Harvard. The platform's goal is to bring next-generation spatial transcriptomics and proteomics approaches to bear on large-scale biological problems. In his parallel research lab, Farhi focuses on continuing technology development and iPS-based screens for understanding psychiatric disease with optical methods. Farhi joined the Broad Institute in July 2018 after completing his Ph.D. in chemical biology at Harvard University. He also holds a B.S. in chemical biology from the University of California, Berkeley.



Stefan Semrau
VP Computational Biology and Data Science,
New York Stem Cell Foundation

I am the Vice President of Computational Biology and Data Science at the New York Stem Cell Foundation (NYSCF) Research Institute and an adjunct Professor of Physics at Leiden University in the Netherlands. My research is focused on the development of computational tools to analyze a large variety of biological data ranging from (single-cell) gene expression to high content screening (HCS). For example, during my PhD research I developed Particle Image (Cross-)Correlation Spectroscopy, a method to obtain nanometer scale correlations from trajectories of single molecules in live cells. A continuation of this work during my postdoctoral training at the Massachusetts Institute of Technology led to a new method for the quantification of expressed gene fusions in individual cancer cells. With my independent group at Leiden University, I continued to work on computational approaches, such as a new measure for the quality of clusters in single-cell data as well as an approach to integrate dynamic transcriptomics and proteomics measurements. At NYSCF one important focus of my team is the analysis of large HCS data sets created by our automation platform, the NYSCF Global Stem Cell Array®.



Johnny Yu
Chief Scientific Officer,
Tahoe Therapeutics

Johnny Yu is a Co-founder and the Chief Scientific Officer at Vevo Therapeutics. He's focused on building and scaling technologies such as Mosaic that allow for high-throughput in vivo drug screening for AI-powered drug discovery. His work sits at the intersection of large-scale single-cell data generation initiatives for drug development across multiple cancer indications. He trained at UCSF with Dr. Hani Goodarzi and Dr. Kevan Shokat and holds a PhD from UCSF. Prior to UCSF, he was at the Broad Institute and Biogen.



Chris Probert
Co-Founder and CTO,
Conflux

Chris Probert is Co-Founder and CTO at Conflux, a software company providing data interoperability and AI infrastructure to support the future of diagnostic medicine. There, Chris oversees development of the software and ML platform, supporting users across clinical labs, biopharma R&D, and research organizations. Previously, Chris led the Clinical Machine Learning team at insitro, a drug discovery company focused on identifying novel drug targets from high-dimensional data. Chris earned a PhD from Stanford, where he developed sequence-to-function models of DNA and applied them to non-invasive biomarker development.



Lukas Fischer
Senior Specialist,
Merck

Data scientist with over 10 years of experience in machine learning and large-scale biological data analysis, specializing in multimodal data integration, computer vision, and spatially resolved biological data. At Merck, I develop models and analysis pipelines that combine imaging and omics data to support drug discovery and biomarker research. Prior to joining Merck, I studied neural population dynamics by integrating behavioral, electrophysiological, and imaging data, using computational modeling to uncover biomarkers of cognition. His doctoral work at the University of Edinburgh focused on building experimental systems to explore neural activity during spatial navigation.



Safiye Celik, PhD
Director of Data Science,
Recursion

Safiye Celik is the Director of Data Science at Recursion, with 18 years of experience in computing and AI, including 12 years leading high-impact initiatives in industry and government research. She holds a PhD in Computer Science and Engineering from the University of Washington, where her research applied machine learning to characterize complex biological systems, uncovering novel insights in cancer and Alzheimer's disease. At Recursion, she leads a team that develops, deploys, and optimizes machine learning models to transform experimental data into comprehensive maps of biology, advancing the vision of industrializing drug discovery.



Angeline Lim
Product Manager,
Molecular Devices

Dr. Angeline Lim is a Product Manager at Molecular Devices, where she drives strategic initiatives and innovation in the high-content imaging portfolio. Formerly a Senior Applications Scientist, Dr. Lim played a pivotal role in developing automated 3d cell culture workflows, imaging and leveraging AI to showcase Molecular Devices products. With over a decade of research experience and a PhD in Molecular Cell and Developmental Biology from UC Santa Cruz, she has authored patents and publications in 3D cell culture and phenotypic analysis.



Nasim Jamali
Product Manager | Director of Morphological Profiling, Ardigen

Nasim Jamali, PhD is a scientist, product leader, and mentor with expertise in morphological profiling, microscopy imaging, and bioimage analysis. She is Director of Morphological Profiling at Ardigen, where she leads product strategy for the phenAID platform, applying AI models to multimodal data to accelerate discovery. Her previous work spans an innovative biotech company, the Broad Institute, and Mount Sinai, with contributions to the JUMP-Cell Painting Consortium and the open-source imaging community through COBA. Nasim earned her PhD at the University of Wisconsin–Madison, where she investigated the role of vitamin D in retinopathy of prematurity. She currently serves as President of the Society for Biomolecular Imaging and Informatics (SBI^2) and co-founded WiSolve Consulting Group.



Emil Rozbicki
President,
Glencoe Software, Inc.

Emil Rozbicki, President at Glencoe Software, is an expert in bioimage data management, visualisation, and analysis at scale. Emil is a physicist by training, prior to joining Glencoe he worked at University of Dundee investigating mechanisms underlying early-stage avian embryo development. During this time, he designed and built the first light sheet microscope in the UK and built analysis routines for investigation of cell behavior at the full organism scale during early-stage development. Currently he is focused on building solutions for the management and analysis of large scale bioimage datasets especially in high content and multiplexed imaging domains for some of the largest academic and biopharma organizations in the world.



Claudia McCown, PhD
Post Doctoral Fellow,
The Herbert Wertheim UF Scripps Institute for Biomedical Innovation & Technology

Claudia earned her Ph.D. in Integrated Biomedical Sciences from the University of Texas Health Science Center at San Antonio in December 2023. Her doctoral research delved into the biophysical and enzymatic properties of the innate immune protein SAMHD1. Currently, Claudia is a Post-Doctoral Fellow in the Spicer/Scampavia lab, where she has been advancing novel high-throughput screening (HTS) methodologies since early 2024. Her current focus is on leveraging induced neuron technologies to enhance neuronal drug discovery processes.



Michael Previte
CTO, SVP R&D, Co-Founder,
Element Biosciences

Michael co-founded Element Biosciences and brings more than 10 years of innovation and discovery in the field of genomics.
Previously, Michael was an Associate Principal Scientist at Illumina. While at Illumina, Michael pioneered innovations in advanced sequencing technologies for both single molecule and ensemble-based methods. He also worked developing high throughput assays for protein evolution and discovery. Prior to Illumina, Michael worked at Life Technologies innovating new single molecule sequencing technologies. During his almost 10 years of work in the sequencing industry, Michael holds four independent sequencing technology patents and has published one of those technologies in Nature Communications.
Michael holds his PhD from Boston College in physical chemistry and received his postdoctoral training from MIT, where he trained in combining advanced microscopy techniques with high throughput assays to study complex biochemical systems. Over his 25 years of work in science and discovery, Michael has over 35 publications and has filed over 25 patents in the areas of plasmonics, sequencing, enzymology, instrumentation, surface chemistry, DNA amplification, high throughput assay discovery, and microscopy.

Main Session Day 2



Thomas Hofstetter
Associate Scientist,
InSphero AG

Thomas Hofstetter is an Associate Scientist at InSphero AG, where he has been developing advanced 3D cell culture models for the past seven years. His research focuses on retinal fibrosis and human brain models, with the goal of building physiologically relevant systems for disease modelling and drug discovery. He is currently pursuing a Master of Advanced Studies in Data Science at the Swiss Distance University of Applied Sciences, complementing his laboratory expertise with computational approaches for multiparametric imaging and assay analysis. His aim is to bridge the gap between wet-lab biology and machine learning, to fully leverage biological data.



Ophir Shalem
Associate Professor,
University of Pennsylvania

Dr. Shalem is an Associate Professor at the Department of Genetics in Perelman School of Medicine at the University of Pennsylvania and the Center for Cellular and Molecular Therapeutics at the Children's Hospital of Philadelphia. He completed his PhD at the Weizmann Institute of Science in Systems Biology. He then moved to the Broad Institute as a postdoctoral fellow in the lab of Prof. Feng Zhang where he was one of the pioneers in the development of CRISPR tools for whole genome functional screens, followed by additional training in the lab of Prof. Andrew Dillin. His lab focuses on the development and applications of functional genomics tools to study cellular proteostasis in health and in neurodegenerative diseases using cell lines and iPSC derived neurons. A major focus area of his lab is the development of scalable pooled protein tagging for both direct manipulation of protein function and multiplexed analysis of protein localization patterns using in situ barcode sequencing and deep learning-driven image analysis. Using these techniques his lab has taken novel unbiased approaches to map the cellular response to spatially restricted induction of protein misfolding, protein recruitment and imaging based unbiased exploration of subcellular proteome dynamics.



Kaylene Simpson
Head, Victorian Centre for Functional Genomics,
Peter MacCallum Cancer Centre

Professor Kaylene Simpson is a leading molecular cell biologist and Head of the Victorian Centre for Functional Genomics (VCFG), established in 2008 at the Peter MacCallum Cancer Centre, Australia. She leads a dynamic team who collaborate widely to deliver high throughput, unbiased target discovery using CRISPR, RNAi, and compound screening across a diversity of models, including primary, patient-derived organoids. The VCFG combines advanced robotics with sophisticated high-content imaging capabilities and provides fully customised analytical pipelines for all projects, enabling rigorous quantitative phenotypic screening and insightful data interpretation. Prof Simpson founded and continues to coordinate the annual Functional High Throughput Technologies conference in Australia and is an original board member and past President of the Society of Biomolecular Imaging and Informatics (SBI2). Under her leadership, the VCFG provides Australian researchers with state-of-the-art high-throughput platforms to drive innovative discovery research.



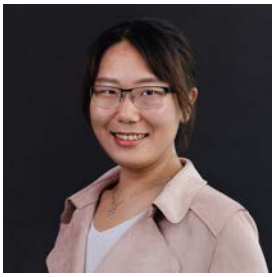
Jaime H. Cheah
Director of Collaborative Screening,
The Broad Institute of MIT and Harvard

Jaime Cheah earned her B.Sc in Biochemistry from McGill University and her Ph.D in Neuroscience from Johns Hopkins School of Medicine. She moved back to the Boston area to do her postdoctoral fellowship at the Novartis Institutes for BioMedical Research and then joined the Broad Institute as a Research Scientist, where she led a team of researchers on a large-scale cancer cell line profiling project, correlating the genetics of cancer to drug responses. She then served as the Director of the High Throughput Sciences Facility at the Koch Institute of MIT for 8 years, where she oversaw a facility outfitted with state-of-the-art equipment and helped scientists design and execute their experiments in a larger, faster and more robust way. In 2022, she returned to the Broad Institute as the Director of Collaborative Screening in the Center for the Development of Therapeutics, where she collaborates with scientists to convert their bench-top research into development of patient therapies, utilizing high throughput drug screening as part of the process. She also is one of the managers of the Broad Institute's Repurposing Hub and connects with a variety of researchers, including those rare disease foundations, to enable access to this valuable resource.



Simon Mathien
Platform Lead, High-Throughput Screening,
Université de Montréal

Dr. Simon Mathien leads the High-Throughput Screening platform at the Institute for Research in Immunology and Cancer (IRIC), Université de Montréal, where he oversees assay development, automation workflows, and large-scale screening campaigns for both academic and industry partners. With a background in molecular and systems biology, and specialized experience in drug discovery and screening technologies, he leads high-throughput screening and early-stage discovery projects that have supported industry collaborations, translational programs, and spin-off creation. Under his leadership, the platform has developed and integrated complex biological models with advanced screening modalities to enable innovative functional genomics and chemical biology programs. Simon's current work focuses on the development and deployment of scalable Live Cell Painting workflows for phenotypic profiling. He co-leads several collaborative projects advancing this emerging technology and exploring its applications in drug discovery research. Beyond his work at IRIC, Simon co-founded and serves on the board of LRIG Canada, a non-profit dedicated to promoting the exchange of knowledge and adoption of automation technologies in biomedical research and drug discovery.



Yue Qin
Postdoctoral Associate,
The Broad Institute of MIT and Harvard

Dr. Yue Qin is a postdoctoral fellow at the Eric and Wendy Schmidt Center of the Broad Institute, co-advised by Drs. Paul Blainey and Caroline Uhler. She is developing computational and experimental methodologies to advance the frontier of optical pooled genetic screens and biological interpretation of cellular images. Dr. Qin received her Bachelor's degree in Bioinformatics from the University of California San Diego (UCSD). She later continued her doctoral training at UCSD in Bioinformatics and Systems Biology, mentored by Dr. Trey Ideker. She was awarded the NCI Predoctoral to Postdoctoral Fellow Transition Award (F99/K00) in 2021. Her thesis work developed machine learning models for integrating different data modalities, and was awarded a Chancellor's Dissertation Medal in 2023. She was named a Siebel Scholar (2022), a Forbes 30 Under 30 Asia (2023), and an Asia-Pacific Leaders Under 30 (2024).



Stephen Straub
Principal Product Specialist,
Revvity

Steve Straub is a Principal Product Specialist for High Content Imaging and Cellular Analysis at Revvity, where he applies his scientific and technical expertise in imaging, image analysis, and automation to support drug discovery and research efforts across a diverse user base. Before joining Revvity, Steve was a Senior Scientist in the Cardiovascular, Metabolic, and Endocrine Diseases Research Group at Pfizer in Groton, CT, where he established an imaging and electrophysiology lab to support the development of novel therapeutics for the treatment of diabetes and obesity. Steve obtained his Ph.D. in Pharmacology from the University of Rochester School of Medicine and performed his postdoctoral training in the lab of Dr. Mark Nelson at the University of Vermont School of Medicine. His academic research focused on elucidating the mechanisms by which intracellular calcium signaling and ion channels control various physiological processes including matching blood flow to neuronal activity in the cerebral cortex, cerebral vascular function, and exocrine cell secretion.



Michael Lippincott
PhD Candidate,
University of Colorado Anschutz Medical Campus

Mike Lippincott is a Computational Cell Biology PhD Candidate in the lab of Gregory Way, PhD, at the University of Colorado Anschutz Medical Campus. Mike's expertise and dissertation focus on leveraging high-dimensional microscopy image data and image-based profiling for a better understanding of cell death dynamics.



Clark Driscoll
Senior Research Scientist,
Curia Global, Inc.

Clark Driscoll is a Senior Research Scientist at Curia's Integrated Drug Discovery Center in Buffalo, NY, where he leads a team of scientists specializing in high-content imaging and high-throughput screening. He and his team design, miniaturize, and optimize a wide array of imaging approaches to address challenging biological questions. Clark has successfully scaled high-content assays from small, bespoke studies to fully automated screens of hundreds of thousands of compounds to supporting both small- and large-molecule discovery programs.



Özlem Yavaş Grining
Senior Scientist,
InSphero AG

Özlem Yavaş Grining is a Senior Scientist in the Technology and Platforms team at InSphero, specializing in high-content imaging and automation. She develops 3D data segmentation and analysis pipelines as well as automated reporting tools to make complex imaging datasets more accessible and reproducible. Her work bridges advanced imaging technologies, data science, and assay development, with a focus on delivering scalable solutions for biotechnology and drug discovery.



Molly McQuilken
Business Development Manager,
Zeiss

Molly McQuilken, is a business development manager for ZEISS. Molly began her career with ZEISS specifically focused on the shaping the ZEISS arivis image analysis platform for biotech and pharma, but has since expanded her focus to full imaging solutions. Before starting her career with ZEISS arivis in 2017, she completed her PhD where she supported the development two new microscopy techniques and of an open-source image analysis toolkit. This experience developing an end-to-end workflow for other microscopists during her PhD, allows her to intuitively understand customer imaging challenges and identify areas where ZEISS can develop solutions to support researchers now and for the long-term.



Kelvin Groot
PhD student
University of Zurich

Kelvin Groot is a PhD student in imaging and systems biology at the University of Zurich. His research focuses on improving image-based drug profiling methods to identify drug mechanisms of action, combining wet lab experimental techniques with computational approaches to enhance our understanding of how therapeutic compounds interact with biological systems. Kelvin completed both his Bachelor's and Master's degrees in Biomedical Sciences at Leiden University in the Netherlands. At the Netherlands Cancer Institute, he worked with Prof. Dr. Rene Bernards on cancer cell senescence and identifying biomarkers of drug response and sensitivity using functional genetic screens. He subsequently joined Prof. Dr. Hans Clevers' laboratory at the Hubrecht Institute, where he characterized a novel human fetal pancreatic organoid model capable of generating cells from all three pancreatic lineages. During his PhD, Kelvin expanded his skill set to include computational biology and image analysis. His current research employs highly multiplexed immunofluorescence imaging to profile oncology drugs, integrating experimental design with machine learning approaches to predict drug mechanisms of action and identify polypharmacological effects. This dual expertise enables him to bridge experimental and computational domains for drug discovery applications.



Marcus Chin
Senior Scientist, Microscopy and High-Content Imaging, Denali Therapeutics

Marcus Chin, PhD, is a Senior Scientist at Denali Therapeutics, where he develops advanced high-content imaging platforms supporting multiple drug discovery and preclinical stage programs in neurodegenerative disease. Since joining Denali in 2022, he has been a core member of the Advanced Bioimaging group and serves as a key contributor biology pathway teams focused on lysosomal biology, neuroimmunology, blood brain barrier biology. Prior to industry, he obtained his PhD in Pharmaceutical Sciences and Pharmacogenomics at UCSF, where he developed image-based high throughput approaches to enhance lysosomal function in neurological cell models. Marcus has led impactful collaborations and authored high-impact publications across various institutions including the UCSF Memory and Aging Center, UCSF Small Molecule Discovery Center, and Gladstone Institutes. He holds a B.A. in biology from Cornell University and a M.Phil. in medical science from the University of Cambridge, where he studied molecular mechanisms of autophagy and cellular homeostasis. Marcus is passionate about integrating cell biology, organellar dynamics, high-throughput assays, and microscopy to drive therapeutic innovation and biological discovery.



Jiří Reiniš
PhD student,
CeMM

Jiří Reiniš studied bioinformatics in Prague and Bologna. In 2021, he joined the lab of Stefan Kubicek at the CeMM Research Center for Molecular Medicine of the Austrian Academy of Sciences in Vienna as a PhD student. In his project, he has developed a computational workflow for the analysis of pooled optical screens for drug-induced changes in subcellular protein localization.

Poster and Speaker Information

Poster titles, abstracts, and placement numbers accessible below.	
SBI2 2025 Posters:	SBI2 2025 Posters

SBI2 Board Member Nominations

The Society invites those interested in supporting its Mission to participate in upcoming elections and Board member nominations.	
Feel free to nominate yourself or someone you believe will be a great fit for the board!	SBI2 Nominations 2026

Conference and Exhibit Hall Map

SBI2
October 27-29, 2025

Renaissance
Boston Waterfront Hotel

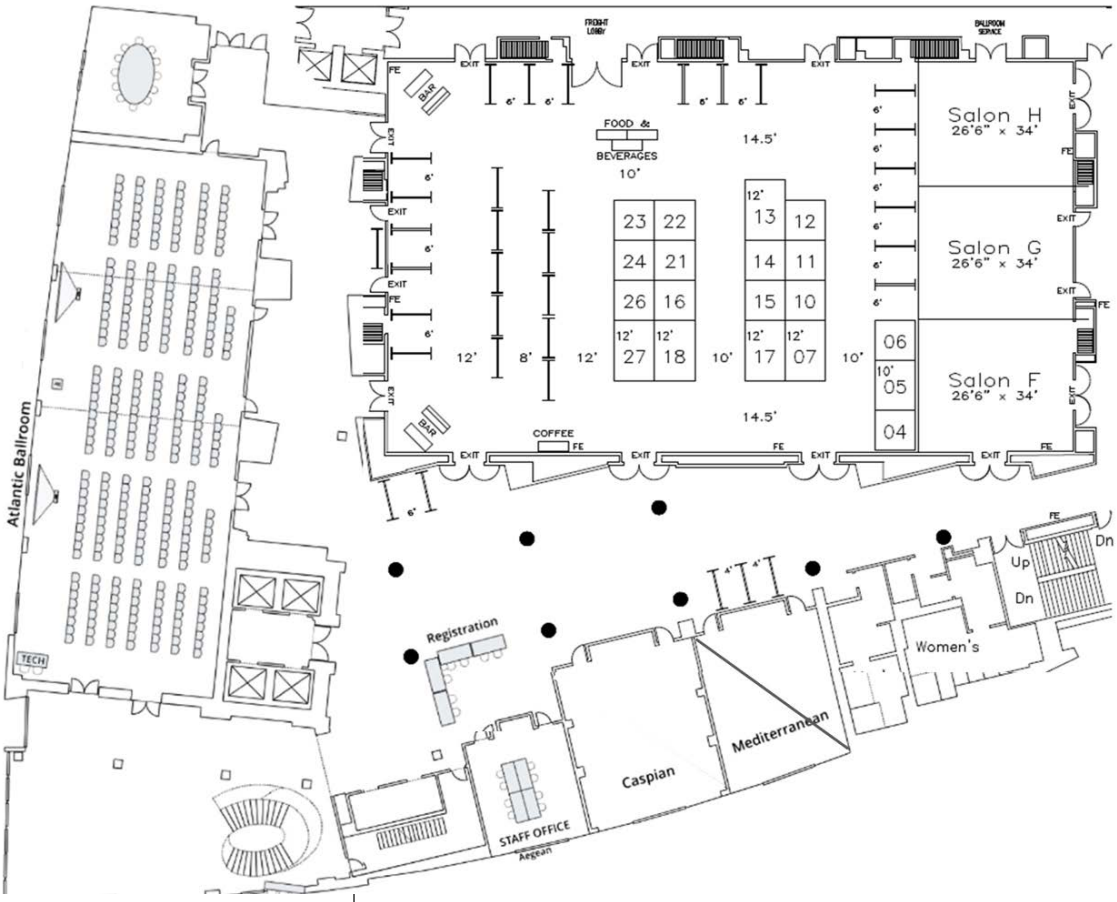


Exhibit Hall and Foyer
• Posters
• Exhibit booths
• High tops + EH map
• Food/Beverage stations
• SBI2 Registration Desk

Exhibitor Booths			
Curia	4	Aurora	16
Ramona Optics	5	Cephla	17
Lumencor	6	Molecular Devices	18
Yokogawa	7	Agilent	21
Mitsui Chemicals	10	Ardigen	22
CytoTronics	11	Genedata	23
Broad Institute	12	Etaluma	24
Araceli Biosciences	13	Saguaro	26
Element Biosciences	14	Revvity	27
Glencoe	15		